RAPHY-AD: RADAR-AWARE FOUNDATION MODEL WITH EXPLICIT PHYSICAL CHAIN-OF-THOUGHT FOR AUTONOMOUS DRIVING

Anonymous authors
Paper under double-blind review

ABSTRACT

Vision-language models (VLMs) have enabled autonomous driving systems to predict trajectories through intermediate reasoning steps. However, VLM-based reasoning is often driven by visual appearance rather than grounded in the physical quantities underlying the scene, such as the distance and relative motion of surrounding agents. Visually driven reasoning can generate linguistically plausible rationales that still remain inconsistent with the actual physical state of the scene, creating a physical-grounding gap between inferred reasoning and directly measured physical quantities. To bridge this gap, we propose ***RaPhy-AD***, a radar-aware foundation model that extends VLM perception beyond vision by incorporating millimeter-wave radar and generates structured physical rationales grounded in radar measurements of object range, azimuth, and Doppler-based radial velocity. To represent these radar-derived physical quantities, *RaPhy-AD* integrates a Physical-Aware Tokenizer for numerical values and physical units and a Polar-Time Radar Encoder that preserves the temporal, azimuth, and range structure of multi-frame radar observations. We further develop a two-stage training strategy consisting of radar-aware supervised fine-tuning followed by Group Relative Policy Optimization with verifiable physical rewards to first learn radar-grounded physical knowledge and then refine the numerical accuracy of physical reasoning and trajectory prediction. *RaPhy-AD* shows strong trajectory prediction performance across challenging driving scenarios, particularly under visual degradation, highlighting the importance of radar-grounded physical information.

1 INTRODUCTION

Vision-language models (VLMs) (Li et al., 2023; Liu et al., 2023; Yang et al., 2025) have emerged as a promising paradigm in autonomous driving (AD) (Zhou et al., 2026b; Xing et al., 2025; Liao et al., 2025), leveraging world knowledge and reasoning capabilities to interpret complex driving scenarios (Shao et al., 2024; Pan et al., 2024; Hegde et al., 2025; Zhou et al., 2026a). Recent VLM-based AD models introduce intermediate reasoning before trajectory generation to bridge scene understanding and motion planning, using textual rationales or structured reasoning chains (Ye et al., 2026; Zhou et al., 2026b). However, these rationales can be linguistically plausible yet physically inconsistent, as they are derived directly from visual observations without explicitly incorporating physical quantities such as object distance and relative motion. This limitation is closely tied to the physical information directly available from the input sensing modalities.

Recent AD research has increasingly incorporated additional sensing modalities, such as LiDAR and millimeter-wave radar, to complement visual perception (Man et al., 2023; Lin et al., 2024). As illustrated in Fig. 1(a), cameras provide rich semantic information, but metric distance and relative motion must be inferred from temporal changes across discrete frames. LiDAR provides explicit metric geometry and

Reviewers: please read the Reviewer Guidelines (iclr.cc/Conferences/2027/ReviewerGuidelines) and the AI Policy for Reviewers (iclr.cc/Conferences/2027/AIPolicyForReviewers). If you used AI to expand, edit, or polish your review, please provide the input text to the LLM. Better still, consider skipping the LLM and submitting your original text: we, and the authors, are much more interested in your unedited thoughts than in what an LLM has to say. AI-assisted or not, you are putting your name and reputation behind your review: LLM-generated falsehoods, hallucinations or misrepresentations are subject to disciplinary action, which may include desk-rejecting all papers you have authored.

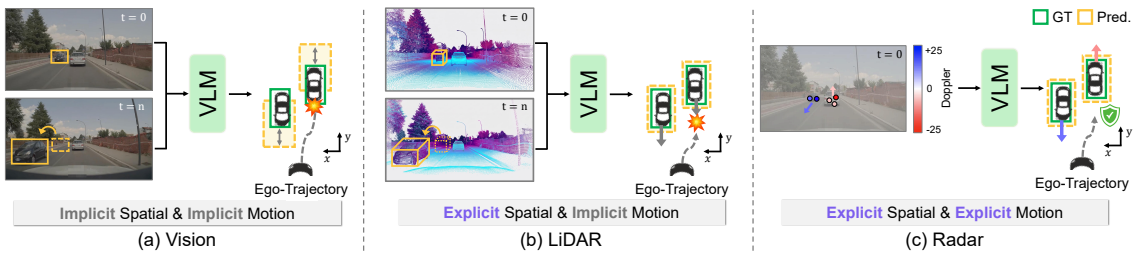


Figure 1: Comparison of implicit and explicit spatial and motion information across vision, LiDAR, and radar for VLM-based autonomous driving and physically grounded trajectory prediction.

range measurements, as shown in Fig. 1(b), yet object velocity still typically requires association across consecutive observations (Li et al., 2026a; Liu et al., 2026). In contrast, radar explicitly provides range and Doppler-based radial velocity within each observation, as depicted in Fig. 1(c). The sensing characteristics of radar have driven the increasing adoption of radar in AI applications (Haitman & Bialer, 2025; Guan et al., 2025; Shin et al., 2026), particularly in AD, where accurate spatial and motion information is essential for physically grounded reasoning and trajectory planning. Despite its unique ability to provide direct spatial and motion measurements, radar has been overlooked in VLM-based autonomous driving.

To bring radar-derived physical cues into VLM-based autonomous driving, we propose ***RaPhy-AD***, a radar-aware foundation model that explicitly generates a structured physical rationale from radar-observed physical measurements for trajectory planning. We introduce two key components for representing radar-derived physical information in the VLM: a Physical-Aware Tokenizer and a Polar-Time Radar Encoder. The former represents numerical quantities and physical units as atomic tokens to preserve their physical meaning and relative magnitudes, while the latter converts multi-frame radar point clouds into VLM-compatible tokens while preserving temporal, azimuth, and range structure. We further propose a training strategy for building a radar-aware foundation model as the pretrained VLM has not been trained on radar observations. We first construct radar-aware supervised fine-tuning data that jointly provide language, visual, and radar supervision, allowing the model to learn radar-grounded physical driving knowledge. We then further optimize the model through reinforcement learning with verifiable rewards (RLVR) using Group Relative Policy Optimization (GRPO) to improve the numerical accuracy of physical reasoning and trajectory prediction.

Our contributions are summarized as follows:

- We introduce *RaPhy-AD*, a radar-aware foundation model that generates structured physical rationales grounded in explicit radar-derived physical quantities before trajectory generation.
- We propose a Physical-Aware Tokenizer and a Polar-Time Radar Encoder to represent numerical physical quantities and preserve the temporal and polar structure of sparse radar observations.
- We develop a two-stage training strategy combining radar-aware supervised fine-tuning with GRPO-based reinforcement learning using verifiable physical rewards, improving the numerical accuracy of generated physical quantities as well as trajectory prediction.

2 Related Work

VLM-Based Autonomous Driving Recent advances in VLMs have introduced language-based reasoning into autonomous driving, extending conventional end-to-end AD methods (Shao et al., 2023; Jia et al., 2023; Yuan et al., 2024; Li et al., 2024a; Sima et al., 2025) are largely encoded implicitly within learned representations. Early VLM-based approaches leverage language- or graph-based reasoning to interpret

driving scenes and support decision-making (Xu et al., 2024b; Sima et al., 2024; Tian et al., 2024; Fu et al., 2024). More recent VLM-based methods incorporate explicit rationales and structured reasoning chains to connect scene understanding with driving decisions and trajectory generation (Nie et al., 2024; Xu et al., 2024a; Xing et al., 2025; Ye et al., 2026; Bassole et al., 2026; Zhang et al., 2026; NVIDIA et al., 2026). Building on this direction, SOLVE (Chen et al., 2025) introduces trajectory chain-of-thought reasoning, while Drive-R1 (Li et al., 2026b) and AutoDrive-R² (Yuan et al., 2026) further align reasoning with motion planning through reinforcement learning. Despite these advances, existing VLM-based approaches remain primarily grounded in visual observations and lack explicit access to physical quantities during reasoning.

Beyond Vision: Physical Sensing for Autonomous Driving To provide explicit physical evidence, a parallel line of work uses geometric cues, including LiDAR point clouds and depth-derived 3D coordinates, for physically grounded spatial reasoning (Wei et al., 2026; Li et al., 2026a; Mandalika et al., 2025). Radar has been extensively studied in autonomous driving for its spatial and motion sensing capabilities. Prior work has primarily explored radar fusion for perception, motion prediction, and end-to-end planning (Lin et al., 2024; Man et al., 2023). More recent studies further exploit radar's Doppler measurements and robustness under adverse conditions for weather-robust perception and planning-oriented end-to-end driving (Chae et al., 2024; Wolters et al., 2025). Despite these advantages, explicitly incorporating radar-derived physical measurements into the reasoning process of VLM-based autonomous driving remains largely unexplored.

3 METHOD

3.1 PROBLEM FORMULATION

Given the ego state E, visual observations V, radar point clouds $\mathcal{P}$, and a high-level driving command C, the objective is to generate a structured driving rationale $\mathcal{R}$ together with a future trajectory $\mathrm{Traj} = \{(x_t, y_t)\}_{t=1}^{T}$. When no driving command is provided, we set $C = \varnothing$. We formulate the joint prediction as

$$P(\mathcal{R}, \mathrm{Traj} \mid E, V, \mathcal{P}, C) = P(\mathcal{R} \mid E, V, \mathcal{P}, C) P(\mathrm{Traj} \mid E, V, \mathcal{P}, C, \mathcal{R}). \tag{1}$$

Rather than treating $\mathcal{R}$ as free-form textual reasoning, we formulate it as a structured physical rationale, $\mathcal{R} = \{\mathcal{R}_{\mathrm{ego}}, \mathcal{R}_{\mathrm{obj}}, \mathcal{R}_{\mathrm{vis}}, \mathcal{R}_{\mathrm{drive}}\}$. Here, $\mathcal{R}_{\mathrm{ego}}$ describes the ego-motion state, $\mathcal{R}_{\mathrm{obj}}$ captures the spatial and motion states of collision-relevant objects, $\mathcal{R}_{\mathrm{vis}}$ summarizes complementary visual scene cues, and $\mathcal{R}_{\mathrm{drive}}$ relates these observations to the subsequent driving decision. Accordingly, ***RaPhy-AD*** models the driving process as $(E, V, \mathcal{P}, C) \rightarrow \mathcal{R} \rightarrow \mathrm{Traj}$, where multimodal observations are first converted into a structured physical rationale, followed by the generation of the future ego trajectory. Unlike conventional VLM-based autonomous driving approaches that primarily rely on visual observations, ***RaPhy-AD*** explicitly leverages radar observations during the intermediate physical reasoning process before trajectory generation.

3.2 OVERALL ARCHITECTURE OF RAPHY-AD

Physical-Aware Tokenizer. As shown in Table 1, a standard LLM tokenizer typically represents numerical values at the digit level, which results in physical quantities such as distances, velocities, and angles being fragmented across multiple tokens. This not only increases the sequence length but also fails to explicitly bind numerical values to their physical units. For example, $5m$ and $5deg$ share the same numerical token representation while their units are encoded separately, and $-1.6deg$ is decomposed into $[-], [1], [.], [6], [deg]$ despite

Table 1: Comparison of tokenization strategies for physical quantities.

	$37m$	$12deg$	$-1.6deg/s$
Qwen3-VL	3 7 m	1 2 deg	$-$ 1 . 6 deg $/s$
Atomic Numeric	37 m	12 deg	-1.6 deg $/s$
Unit-Integrated	3 7 m	1 2 deg	$-$ 1 . 6 deg/s
Atomic + Unit	$37m$	$12deg$	$-1.6deg/s$

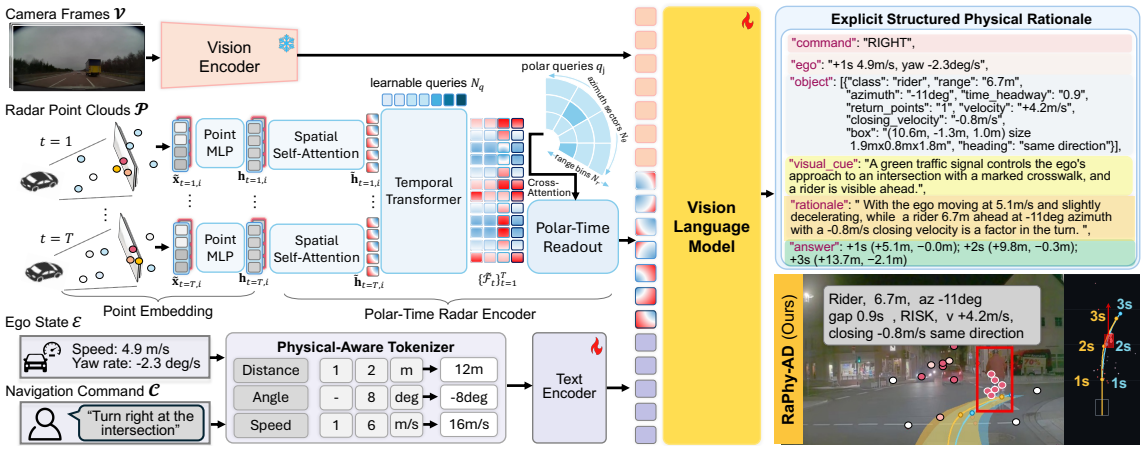


Figure 2: **Overview of RaPhy-AD**. The model integrates a Polar-Time Radar Encoder and Physical-Aware Tokenizer to generate an Explicit Structured Physical Rationale from radar inputs for trajectory prediction.

representing a single physical quantity. The *Atomic Numeric* strategy represents each numerical value as a single token, whereas *Unit-Integrated* represents each compound physical unit as a single token. Our *Atomic + Unit* strategy combines the two, encoding a complete physical quantity such as $37m$, $12deg$, or $7.3m/s$ as a single token. The resulting physics-aware vocabulary covers physical quantities and associated units, including distance, velocity, azimuth, yaw rate, and image-space coordinates. Since these physics-aware tokens are newly added to the pretrained vocabulary, their embeddings must also be initialized. Rather than initializing them randomly, we use a structured initialization that reflects both unit consistency and numerical proximity. Tokens sharing the same physical unit are initialized similarly, while neighboring numerical values are placed close in the embedding space, providing a more stable starting point for training.

Point Embedding. As illustrated in Fig. 2, we introduce a radar encoder to incorporate radar observations into the VLM by converting raw radar point clouds into a sequence of radar tokens. The radar encoder takes a sequence of consecutive radar frames $\mathcal{P} = \{\mathcal{P}_t\}_{t=1}^{T}$, where each frame is represented as a variable-sized point set $\mathcal{P}_t = \{p_{t,i}\}_{i=1}^{M_t}$ and M_t denotes the number of radar points in frame t. The encoder transforms these sparse point clouds into a fixed-length sequence of radar tokens while preserving spatial and temporal information. For each radar point p_i, we use an 8-dimensional point representation

$$\mathbf{x}_i = [x_i, y_i, r_i, \sin\theta_i, \cos\theta_i, v_i^{\mathrm{rad}}, \Delta v_i^{\mathrm{dop}}, \rho_i], \tag{2}$$

where (x_i, y_i) denote the Cartesian coordinates, r_i and θ_i are the range and azimuth, v_i^{rad} is the radial velocity, $\Delta v_i^{\mathrm{dop}}$ is the ego-motion-compensated Doppler residual, and ρ_i denotes the radar cross section (RCS) of the i-th point scatterer. Among these features, (x_i, y_i, r_i) directly encode the spatial location and range of each radar point. To better represent their relative spatial differences, we augment each geometric coordinate $u \in \{x_i, y_i, r_i\}$ with a sinusoidal Fourier encoding defined as

$$\phi(u) = [\sin(2^0\pi u), \cos(2^0\pi u), \ldots, \sin(2^{K-1}\pi u), \cos(2^{K-1}\pi u)] \in \mathbb{R}^{2K}, \tag{3}$$

where K denotes the number of Fourier frequencies. The Fourier features of (x_i, y_i, r_i) are concatenated with the point representation $\mathbf{x}_i$ and projected into the radar point embedding space:

$$\tilde{\mathbf{x}}_{t,i} = [\mathbf{x}_{t,i}, \phi(x_{t,i}), \phi(y_{t,i}), \phi(r_{t,i})] \in \mathbb{R}^{8+6K}, \qquad \mathbf{h}_{t,i} = f_{\mathrm{pt}}(\tilde{\mathbf{x}}_{t,i}) \in \mathbb{R}^{D}, \tag{4}$$

where f_{pt} denotes the point-wise projection MLP, D denotes the point embedding dimension, and $\mathbf{h}_{t,i}$ is the embedding of radar point i in frame t. The point embeddings in each frame t are represented as $\mathcal{H}_t = \{\mathbf{h}_{t,i}\}_{i=1}^{M_t}$. Additional implementation details are provided in Appendix D.

Polar-Time Radar Encoder. The point embeddings $\mathcal{H}_t$ are then processed by geometrical self-attention to capture interactions among radar points within the same frame, resulting in spatially contextualized representations $\widetilde{\mathcal{H}}_t$. Since the number of radar points M_t varies across frames and can be too large for direct input to the language model, we use N_q learnable queries to aggregate $\widetilde{\mathcal{H}}_t$ through cross-attention into a fixed-length set of frame tokens $\mathcal{F}_t = \{\mathbf{f}_{t,n}\}_{n=1}^{N_q}$. The resulting frame tokens $\{\mathcal{F}_t\}_{t=1}^{T}$, augmented with temporal positional embeddings, are then processed by a temporal Transformer to model cross-frame dependencies, yielding temporally contextualized representations $\{\widetilde{\mathcal{F}}_t\}_{t=1}^{T}$. Here, considering the intrinsic polar nature of radar measurements, a *polar-time readout* is applied to each frame to preserve explicit temporal, azimuth, and range structure in the radar tokens. Namely, to explicitly encode the spatial structure, each frame is partitioned into N_θ azimuth sectors and N_r range bins, resulting in $N_\theta N_r$ polar regions. Each region j is associated with a learnable query $\mathbf{q}_j$ and a spatial anchor $(\bar{\theta}_j, \bar{r}_j)$ defined by its azimuth and range center. For a query $\mathbf{q}_j$ centered at $(\bar{\theta}_j, \bar{r}_j)$, the attention weight and the corresponding radar token are computed as

$$a_{ji} = \mathrm{softmax}_i\left(\frac{\mathbf{q}_j^\top \mathbf{k}_i}{\sqrt{D}} - \alpha_\theta|\theta_i - \bar{\theta}_j| - \alpha_r|r_i - \bar{r}_j|\right), \qquad \mathbf{z}_j = \sum_i a_{ji}\mathbf{v}_i \in \mathbb{R}^D. \tag{5}$$

The spatial bias guides each query toward radar returns near its assigned azimuth–range region while allowing soft attention to neighboring regions. Each token therefore represents a specific frame and polar region, preserving both spatial and motion information. Collecting the readout tokens across all T frames yields the radar token sequence $\mathbf{Z} \in \mathbb{R}^{N_{\text{rad}} \times D}$, where $N_{\text{rad}} = TN_\theta N_r$ denotes the total number of radar tokens.

Radar-Vision-Language Model Integration. *RaPhy-AD* integrates the proposed *physical-aware tokenizer* and *Polar-Time Radar Encoder* into a pretrained VLM to enable joint reasoning over visual, radar, and ego-motion information. The visual observations V are processed by the native vision encoder of the pretrained VLM, producing visual tokens. In parallel, the radar point clouds $\mathcal{P}$ are encoded by the proposed *Polar-Time Radar Encoder* into radar tokens. The ego-motion history E, optional high-level driving command C, and task instruction are all processed by our *physical-aware tokenizer*, where E is represented using predefined axis-specific trajectory tokens and the remaining textual inputs are encoded as standard text tokens. The resulting multimodal tokens are jointly processed by the pretrained VLM backbone to generate task-specific outputs. For structured driving reasoning, the model generates the physical rationale $\mathcal{R}$ followed by the future trajectory Traj, following $(E, V, \mathcal{P}, C) \rightarrow \mathcal{R} \rightarrow \mathrm{Traj}$. Importantly, ***RaPhy-AD*** employs no task-specific prediction heads; different driving tasks are specified through language instructions and generated autoregressively by VLM. Additional implementation details are provided in Appendix D

3.3 Training Strategy

Radar-Aware Adaptation with Supervised Fine-Tuning Building upon a pretrained VLM backbone with strong visual understanding, we first perform supervised fine-tuning (SFT) to adapt the model for radar-aware perception and physical reasoning in autonomous driving. Since the pretrained VLM has not been exposed to radar observations, this stage performs SFT using a diverse set of autonomous-driving instruction–response pairs automatically constructed from the NVIDIA PhysicalAI-Autonomous Vehicles dataset (NVIDIA et al., 2026), enabling the model to integrate radar representations with the VLM backbone and develop physical reasoning capabilities. By training across diverse downstream driving tasks, the model learns radar-grounded geometric and motion representations that support structured textual reasoning about physical states. Through this supervision, the Physical-Aware Tokenizer, the Polar-Time Radar Encoder,

and the VLM backbone are jointly adapted. To encourage the model to effectively utilize both visual and radar cues, we further employ modality dropout during training by randomly dropping either the camera or radar input. Further details on the construction of the SFT dataset are provided in Appendix B.1, while the training setup, hyperparameters, and other implementation details are described in Appendix C.

Verifiable Reward Optimization for Physical Reasoning. Although SFT enables the model to acquire radar-aware physical knowledge, it primarily optimizes the likelihood of target responses rather than the numerical accuracy of predicted physical quantities, such as object range, velocity, ego motion, and trajectory. Consequently, these predictions are not explicitly optimized according to their numerical deviation from the ground truth. We therefore further optimize *RaPhy-AD* using RLVR, where verifiable rewards are computed by comparing the generated ego and object states, visual cues, driving rationale, and trajectory against their corresponding ground-truth annotations. For each multimodal prompt, we sample a group of responses and use their normalized rewards to estimate relative advantages, which are then used for GRPO without a separate value network. RLVR provides explicit feedback on the numerical accuracy and consistency of physical reasoning and trajectory prediction without relying on a learned reward model or human preference labels.

Verifiable Physical Reward Design. We design deterministic verifiable rewards to directly evaluate the physical correctness of the structured rationale $\mathcal{R}$ and the final trajectory Traj. Specifically, the generated rationale components $\mathcal{R}_{\mathrm{ego}}$, $\mathcal{R}_{\mathrm{obj}}$, $\mathcal{R}_{\mathrm{vis}}$, and $\mathcal{R}_{\mathrm{drive}}$ are evaluated against their corresponding ground-truth fields constructed as described in Appendix B.1, while Traj is evaluated against the ground-truth ego trajectory. For numerical physical quantities, we map a nonnegative prediction error e to a bounded reward $d(e;h)$, where $h>0$ is a quantity-specific tolerance. For the ego state, v_{ego} and $\dot{\psi}_{\mathrm{ego}}$ denote the ground-truth ego speed and yaw rate, respectively, while $\hat{v}_{\mathrm{ego}}$ and $\hat{\dot{\psi}}_{\mathrm{ego}}$ denote their predicted values. The ego-state reward is computed as the average of the corresponding speed and yaw-rate scores:

$$d(e;h)=\frac{h}{h+e}, \qquad R_{\mathrm{ego}}=\frac{1}{2}\left[d(|\hat{v}_{\mathrm{ego}}-v_{\mathrm{ego}}|;h_v)+d\Big(|\hat{\dot{\psi}}_{\mathrm{ego}}-\dot{\psi}_{\mathrm{ego}}|;h_{\dot{\psi}}\Big)\right]. \tag{6}$$

For object-level reasoning, predicted collision-relevant objects are matched to ground-truth objects based on spatial proximity. For each matched pair i, the object-state reward measures the agreement between the predicted and ground-truth range r_i, azimuth θ_i, radar-point count n_i^{radar}, and closing velocity v_i^{close}:

$$R_{\mathrm{obj}}^{(i)}=\frac{1}{4}\Big[d(|\hat{r}_i-r_i|\,;h_r)+d\Big(\Big|\hat{\theta}_i-\theta_i\Big|\,;h_\theta\Big)+d\big(\big|\hat{n}_i^{\mathrm{radar}}-n_i^{\mathrm{radar}}\big|\,;h_n\big)+d\big(\big|\hat{v}_i^{\mathrm{close}}-v_i^{\mathrm{close}}\big|\,;h_{\mathrm{close}}\big)\Big]. \tag{7}$$

The resulting object reward aggregates these scores over spatially matched objects while also accounting for missed or spurious objects. The visual-cue reward verifies driving-relevant scene elements represented in $\mathcal{R}_{\mathrm{vis}}$, such as traffic signals, stop lines, crosswalks, and lane constraints. The trajectory reward measures the geometric agreement between the predicted and ground-truth ego trajectories. For $\mathcal{R}_{\mathrm{drive}}$, we evaluate whether the synthesized driving decision is consistent with the preceding ego, object, and visual states and the corresponding ground-truth rationale. Detailed implementation settings are provided in Appendix E.

4 Experiments

4.1 Main Results

We first evaluate the trajectory prediction performance of RaPhy-AD against prior VLM-based autonomous driving models (Zhou et al., 2026b; Xing et al., 2025; Yuan et al., 2026; NVIDIA et al., 2026; Shi et al., 2026; Zhao et al., 2026). Table 2 reports L2 trajectory error and collision rate at 1 s, 2 s, and 3 s prediction horizons. All models are evaluated on 1,287 trajectory anchors from 1,210 held-out test clips in

Table 2: Comparison of trajectory prediction (L2, m) and collision rate (%) on the NVIDIA PhysicalAI-AV dataset. C and R denote camera and radar, respectively. Bold indicates the best performance.

Method	Sensors	L2 (m) ↓				Collision Rate (%) ↓			
		1s	2s	3s	Avg.	1s	2s	3s	Avg.
Prior VLM/VLA Models (Trained on nuScenes)									
AutoVLA	3×C	4.166	7.090	10.142	7.133	0.894	1.845	3.030	1.923
OpenEMMA	1×C	1.849	3.113	4.415	3.126	10.566	1.901	3.654	2.041
AutoDrive-R^2	1×C	0.817	1.465	2.276	1.520	0.233	0.583	1.049	0.622
Prior VLM/VLA Models (Trained on NVIDIA PhysicalAI-AV)									
Alpamayo 1	4×C	0.088	0.261	0.527	0.292	0.000	0.000	0.045	0.015
Alpamayo 1.5	4×C	0.091	0.303	0.653	0.349	0.000	0.010	0.130	0.046
DriveWAM	1×C	0.119	0.305	0.602	0.342	0.000	0.058	0.207	0.088
SimWAM	1×C	0.155	0.295	0.556	0.335	0.000	0.000	0.104	0.035
RaPhy-AD (Trained on NVIDIA PhysicalAI-AV, Ours)									
SFT (w/o Rationale.)	1×C+R	0.093	0.263	0.544	0.300	0.000	0.039	0.181	0.073
SFT (w/ Rationale.)	1×C+R	0.103	0.238	0.466	0.269	0.000	0.019	0.130	0.050
SFT + GRPO	1×C+R	**0.072**	**0.126**	**0.236**	**0.144**	**0.000**	**0.000**	**0.042**	**0.014**

Table 3: Zero-shot evaluation on nuScenes.

Method	L2 (m) ↓				Collision (%) ↓			
	1s	2s	3s	Avg.	1s	2s	3s	Avg.
Prior Methods								
ST-P3	1.330	2.110	2.900	2.110	0.230	0.620	1.270	0.710
UniAD	0.450	0.700	1.040	0.730	0.620	0.580	0.630	0.610
OccNet	1.290	2.130	2.990	2.140	0.210	0.590	1.370	0.720
VAD	0.410	0.700	1.050	0.720	0.070	0.170	0.410	0.220
SparseDrive	0.290	0.580	0.960	0.610	0.010	0.050	0.180	0.080
DiffusionDrive	0.270	0.540	0.900	0.570	0.030	0.050	0.160	0.080
SimWAM	0.290	0.820	1.770	0.960	**0.000**	**0.030**	**0.110**	**0.050**
RaPhy-AD (Ours)								
SFT (w/o Rat.)	0.188	0.458	0.841	0.496	0.01	0.064	0.209	0.071
SFT (w/ Rat.)	0.160	0.401	0.771	0.443	0.020	0.044	0.144	0.069
SFT + GRPO	**0.157**	**0.385**	**0.722**	**0.422**	0.010	0.039	0.166	0.072

Table 4: Trajectory Prediction across Driving Scenarios.

Scenario	Alpamayo 1.5				Ours			
	1s	2s	3s	Avg.	1s	2s	3s	Avg.
Ego Behavior								
Lane Keeping	0.091	0.281	0.603	0.325	0.061	0.118	0.220	0.132
Lane Change	0.101	0.324	0.668	0.365	0.079	0.141	0.287	0.169
Start After Stop	0.093	0.331	0.771	0.397	0.068	0.144	0.331	0.178
Interaction								
Pedestrian/Cyclist	0.102	0.353	0.751	0.401	0.073	0.131	0.249	0.148
Cut-in	0.111	0.371	0.768	0.418	0.079	0.15	0.273	0.168
Dense Traffic	0.094	0.323	0.694	0.366	0.071	0.121	0.232	0.141
Environment								
Snow	0.101	0.311	0.672	0.358	0.071	0.132	0.254	0.149
Fog	0.107	0.315	0.607	0.345	0.071	0.141	0.213	0.141
Night	0.093	0.291	0.628	0.336	0.071	0.123	0.224	0.135

the NVIDIA PhysicalAI-AV dataset (NVIDIA et al., 2026). For completeness, we report models trained on both nuScenes and NVIDIA PhysicalAI-AV, all evaluated on the held-out NVIDIA PhysicalAI-AV test set. Since many of prior VLM/VLA driving models are trained on nuScenes, we include them as reference points, while PhysicalAI-AV-trained models serve as the primary comparison. Among prior models trained on NVIDIA PhysicalAI-AV, Alpamayo 1 achieves the lowest average L2 error of 0.292 m, whereas our SFT+GRPO model reduces it to 0.144 m, corresponding to a 50.7% reduction. *RaPhy-AD* shows increasingly pronounced improvements at longer prediction horizons, particularly at 3s, while maintaining consistently low collision rates. Incorporating structured rationale supervision into SFT already improves trajectory prediction, reducing the average L2 error from 0.300 m to 0.269 m. Further optimization with RLVR using GRPO substantially lowers the error to 0.144 m, suggesting that verifiable reinforcement learning is particularly effective for improving numerical physical reasoning beyond supervised training alone.

Table 3 further evaluates the zero-shot generalization of *RaPhy-AD* on nuScenes, without additional training on the target dataset. Despite the domain shift, *RaPhy-AD* generalizes favorably to the unseen dataset: SFT+GRPO achieves an average L2 error of 0.42 m, outperforming the strongest reported baseline in the table, DiffusionDrive (0.57,m), by 26.3%. Structured rationale supervision also improves zero-shot performance, reducing the average L2 error from 0.48 m without rationale to 0.44 m with rationale, while GRPO

Table 5: Effect of radar inputs on trajectory prediction.

Model	1s	2s	3s	Avg.
Zero Radar				
SFT w/o rationale	0.118	0.358	0.768	0.415
SFT w/ rationale	0.113	0.276	0.553	0.314
SFT + GRPO	0.119	0.285	0.566	0.323
Random Radar				
SFT w/o rationale	0.099	0.289	0.611	0.333
SFT w/ rationale	0.196	0.436	0.827	0.486
SFT + GRPO	0.159	0.401	0.791	0.450
Original Radar				
SFT w/o rationale	0.093	0.263	0.544	0.300
SFT w/ rationale	0.103	0.238	0.466	0.269
SFT + GRPO	**0.072**	**0.126**	**0.236**	**0.144**

Table 6: Ablation Study of Tokenization and Radar Encoding.

Variant	1s	2s	3s	Avg.
Tokenizer				
Standard	0.328	1.163	2.655	1.382
Atomic Numeric	0.239	0.989	2.418	1.215
Unit-Integrated	0.240	0.988	2.412	1.214
Atomic + Unit	**0.214**	**0.915**	**2.277**	**1.135**
Radar Encoder (w/ Atomic + Unit)				
Global	0.298	1.102	2.585	1.328
Frame	0.235	0.970	2.367	1.191
Polar	0.246	0.997	2.399	1.214
Frame + Polar	**0.225**	**0.957**	**2.346**	**1.176**

further reduces it to 0.42 m. Although SimWAM obtains the lowest average collision rate of 0.05%, *RaPhy-AD* remains competitive at 0.07%, while achieving stronger trajectory accuracy. These results suggest that structured physical reasoning grounded in radar-derived spatial and motion information, such as object range and relative velocity, remains effective even in zero-shot environments.

4.2 Ablation Studies

Across Diverse Scenarios: RaPhy-AD vs. VLM. To investigate when radar-derived physical information is particularly beneficial, we compare *RaPhy-AD* with Alpamayo 1.5, a representative vision-based driving model, across diverse driving scenarios (Table 4). The evaluation covers nine scenarios across three groups: ego behaviors, interaction-heavy situations, and adverse environmental conditions. *RaPhy-AD* consistently outperforms Alpamayo 1.5 across these scenarios, with particularly clear improvements under adverse environmental conditions. In snow, fog, and night, *RaPhy-AD* achieves average L2 errors of 0.149 m 0.141 m and 0.135 m, respectively, compared with 0.358 m, 0.345 m and 0.336 m for Alpamayo 1.5. These improvements suggest that radar-derived physical cues, particularly explicit range and relative-motion information, can complement visual perception when appearance or visibility becomes unreliable, leading to more consistent trajectory prediction under challenging environmental conditions.

Effect of Radar. To assess the extent to which *RaPhy-AD* leverages radar information for trajectory prediction, we evaluate the model under three radar conditions: zero radar, random radar from other clips, and original radar, as summarized in Table 5. When radar is removed, its average L2 error increases from 0.144 m to 0.323 m ($2.2\times$), and replacing the radar input with mismatched radar from other clips further increases the error to 0.450 m ($3.1\times$). In contrast, SFT without rationale degrades more moderately from 0.300 m to 0.415 m ($1.4\times$) under zero radar, suggesting that GRPO encourages the model to exploit radar-derived physical cues more strongly for trajectory generation. Notably, random radar is more harmful than zero radar for most configurations: zero radar allows the model to fall back on visual information, whereas mismatched radar provides plausible but incorrect physical cues that can actively mislead the prediction.

Model Components. We conduct ablation studies on the Physical-Aware Tokenizer and the Polar-Time Radar Encoder to examine how effectively these components represent radar-derived physical information (Table 6). For a fair and efficient controlled comparison, all variants are trained for 100 optimization steps under the same setting. For tokenization, Atomic Numeric and Unit-Integrated designs improve trajectory prediction over the standard tokenizer by directly representing continuous physical values and preserving their associated physical units, reducing the average L2 error from 1.382 m to 1.215 m and 1.214 m, respectively. The Atomic + Unit design achieves the best performance with an average L2 error of 1.135 m,

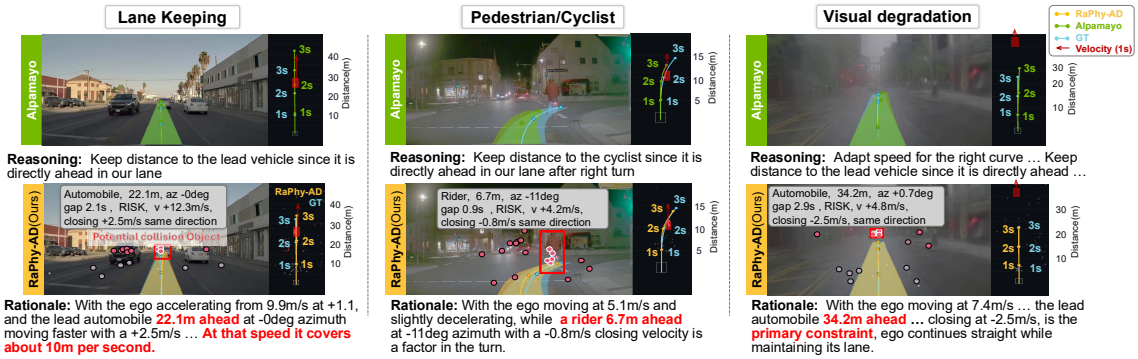


Figure 3: Qualitative comparison of *RaPhy-AD* and Alpamayo 1.5 across diverse driving scenarios.

indicating that numerical values and their physical units provide complementary information for representing radar-derived quantities. For radar encoding, both frame-wise and polar representations outperform global encoding by preserving the temporal organization of radar observations and their native geometric structure, respectively. Combining both in the Frame+Polar encoder yields the lowest average L2 error of 1.176 m, indicating that jointly modeling temporal and radar-specific geometric structure is more effective for trajectory prediction under the controlled training budget and limited optimization steps.

4.3 Qualitative Results

As shown in Fig. 3, we qualitatively compare *RaPhy-AD* with Alpamayo across lane-keeping, pedestrian/cyclist, and visual-degradation scenarios, focusing on how each model interprets scene context. In the lane-keeping scenario, Alpamayo fails to accurately estimate the distance to the lead vehicle, resulting in a trajectory deviation of more than $10\,\mathrm{m}$, whereas *RaPhy-AD* explicitly reasons about the collision-relevant vehicle as being 22.1 m ahead with a +2.5 m/s relative motion, and generates a trajectory that more closely follows the ground truth. In the pedestrian/cyclist scenario, *RaPhy-AD* identifies the cyclist as a collision-relevant object 6.7 m ahead and incorporates its $-0.8\,\mathrm{m/s}$ closing velocity into the driving rationale, allowing the model to account for the interaction when planning the turn. Under visual degradation, radar-derived object states remain available despite weakened visual evidence, enabling *RaPhy-AD* to identify the lead vehicle 34.2 m ahead with a $-2.5\,\mathrm{m/s}$ closing velocity and maintain a trajectory consistent with the driving context. These results highlight the benefit of radar-derived physical cues for spatial reasoning, interaction-aware planning, and robust trajectory prediction under degraded visual conditions.

5 Conclusion

We introduced *RaPhy-AD*, a radar-aware foundation model that grounds autonomous driving reasoning in radar-derived physical measurements. By integrating a Physical-Aware Tokenizer and Polar-Time Radar Encoder into a pretrained VLM, RaPhy-AD generates structured physical rationales before trajectory prediction, explicitly connecting scene understanding, physical reasoning, and driving decisions. A two-stage training strategy combining radar-aware SFT and GRPO-based RLVR further improves the numerical fidelity of both physical reasoning and trajectory generation. Experiments and ablations demonstrate the effectiveness of radar observations, structured rationale supervision, and radar-specific representation design, with particular benefits in interaction-heavy and visually degraded scenarios. These results suggest that explicit physical rationales provide an effective intermediate representation for bridging multimodal scene understanding and physically grounded trajectory generation under challenging conditions.

6 AI USE STATEMENT

Generative AI tools were used solely for grammar correction and typographical error checking. All scientific content, methodology, experiments, analyses, and conclusions were developed and verified by the authors.

REFERENCES

Yipene Cedric Francois Bassole, Sungwoo Kim, Jiwoo Jung, and Yunsick Sung. Hybriddrivevla: Vision-language-action model with visual cot reasoning and tot evaluation for autonomous driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 32421–32430, 2026.

Yujeong Chae, Hyeonseong Kim, and Kuk-Jin Yoon. Towards robust 3d object detection with lidar and 4d radar fusion in various weather conditions. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 15162–15172, 2024. doi: 10.1109/CVPR52733.2024.01436.

Xuesong Chen, Linjiang Huang, Tao Ma, Rongyao Fang, Shaoshuai Shi, and Hongsheng Li. Solve: Synergy of language-vision and end-to-end networks for autonomous driving. In *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 12068–12077. IEEE, 2025.

Jie Cheng, Yingbing Chen, and Qifeng Chen. Pluto: Pushing the limit of imitation learning-based planning for autonomous driving. *arXiv preprint arXiv:2404.14327*, 2024a.

Jie Cheng, Yingbing Chen, Xiaodong Mei, Bowen Yang, Bo Li, and Ming Liu. Rethinking imitation-based planners for autonomous driving. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 14123–14130. IEEE, 2024b.

Daocheng Fu, Xin Li, Licheng Wen, Min Dou, Pinlong Cai, Botian Shi, and Yu Qiao. Drive like a human: Rethinking autonomous driving with large language models. In *2024 IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)*, pp. 910–919. IEEE, 2024.

Runwei Guan, Ruixiao Zhang, Ningwei Ouyang, Jianan Liu, Ka Lok Man, Xiaohao Cai, Ming Xu, Jeremy Smith, Eng Gee Lim, Yutao Yue, and Hui Xiong. Talk2radar: Bridging natural language with 4d mmwave radar for 3d referring expression comprehension. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 10884–10891, 2025. doi: 10.1109/ICRA55743.2025.11128399.

Yuval Haitman and Oded Bialer. Doppdrive: Doppler-driven temporal aggregation for improved radar object detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 26085–26094, October 2025.

Deepti Hegde, Rajeev Yasarla, Hong Cai, Shizhong Han, Apratim Bhattacharyya, Shweta Mahajan, Litian Liu, Risheek Garrepalli, Vishal M. Patel, and Fatih Porikli. Distilling multi-modal large language models for autonomous driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 27575–27585, June 2025.

Shengchao Hu, Li Chen, Penghao Wu, Hongyang Li, Junchi Yan, and Dacheng Tao. St-p3: End-to-end vision-based autonomous driving via spatial-temporal feature learning. In *European Conference on Computer Vision*, pp. 533–549. Springer, 2022.

Yihan Hu, Jiazhi Yang, Li Chen, Keyu Li, Chonghao Sima, Xizhou Zhu, Siqi Chai, Senyao Du, Tianwei Lin, Wenhai Wang, et al. Planning-oriented autonomous driving. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 17853–17862. IEEE, 2023.

Xiaosong Jia, Penghao Wu, Li Chen, Jiangwei Xie, Conghui He, Junchi Yan, and Hongyang Li. Think twice before driving: Towards scalable decoders for end-to-end autonomous driving. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 21983–21994. IEEE, 2023.

Bo Jiang, Shaoyu Chen, Qing Xu, Bencheng Liao, Jiajie Chen, Helong Zhou, Qian Zhang, Wenyu Liu, Chang Huang, and Xinggang Wang. Vad: Vectorized scene representation for efficient autonomous driving. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 8306–8316. IEEE, 2023.

Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett (eds.), *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 19730–19742. PMLR, 23–29 Jul 2023. URL https://proceedings.mlr.press/v202/li23q.html.

Peizheng Li, Zhenghao Zhang, David Holtz, Hang Yu, Yutong Yang, Yuzhi Lai, Rui Song, Andreas Geiger, and Andreas Zell. Spacedrive: Infusing spatial awareness into vlm-based autonomous driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 40096–40107, June 2026a.

Yue Li, Meng Tian, Dechang Zhu, Jiangtong Zhu, Zhenyu Lin, Zhiwei Xiong, and Xinhai Zhao. Drive-r1: Bridging reasoning and planning in vlms for autonomous driving with reinforcement learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pp. 6708–6716, 2026b.

Zhenxin Li, Kailin Li, Shihao Wang, Shiyi Lan, Zhiding Yu, Yishen Ji, Zhiqi Li, Ziyue Zhu, Jan Kautz, Zuxuan Wu, et al. Hydra-mdp: End-to-end multimodal planning with multi-target hydra-distillation. *arXiv preprint arXiv:2406.06978*, 2024a.

Zhiqi Li, Wenhai Wang, Hongyang Li, Enze Xie, Chonghao Sima, Tong Lu, Qiao Yu, and Jifeng Dai. Bevformer: learning bird's-eye-view representation from lidar-camera via spatiotemporal transformers. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 47(3):2020–2036, 2024b.

Bencheng Liao, Shaoyu Chen, Xinggang Wang, Tianheng Cheng, Qian Zhang, Wenyu Liu, and Chang Huang. Maptr: Structured modeling and learning for online vectorized hd map construction. *arXiv preprint arXiv:2208.14437*, 2022.

Bencheng Liao, Shaoyu Chen, Haoran Yin, Bo Jiang, Cheng Wang, Sixu Yan, Xinbang Zhang, Xiangyu Li, Ying Zhang, Qian Zhang, et al. Diffusiondrive: Truncated diffusion model for end-to-end autonomous driving. In *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 12037–12047. IEEE, 2025.

Zhiwei Lin, Zhe Liu, Zhongyu Xia, Xinhao Wang, Yongtao Wang, Shengxiang Qi, Yang Dong, Nan Dong, Le Zhang, and Ce Zhu. Rcbevdet: Radar-camera fusion in bird's eye view for 3d object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 14928–14937, June 2024.

Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine (eds.), *Advances in Neural Information Processing Systems*, volume 36, pp. 34892–34916. Curran Associates, Inc., 2023. doi: 10.52202/075280-1516. URL https://proceedings.neurips.cc/paper_files/paper/2023/file/6dcf277ea32ce3288914faf369fe6de0-Paper-Conference.pdf.

Yicheng Liu, Jinghuai Zhang, Liangji Fang, Qinhong Jiang, and Bolei Zhou. Multimodal motion prediction with stacked transformers. In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 7573–7582. IEEE, 2021.

Zhe Liu, Runhui Huang, Rui Yang, Siming Yan, Zining Wang, Lu Hou, Di Lin, Xiang Bai, and Hengshuang Zhao. Drivepi: Spatial-aware 4d mllm for unified autonomous driving understanding, perception, prediction and planning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 3688–3698, June 2026.

Yunze Man, Liang-Yan Gui, and Yu-Xiong Wang. Bev-guided multi-modality fusion for driving perception. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 21960–21969, June 2023.

Sriram Mandalika, Athira Nambiar, et al. Primedrive-cot: A precognitive chain-of-thought framework for uncertainty-aware object interaction in driving scene scenario. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 5332–5340, 2025.

Ming Nie, Renyuan Peng, Chunwei Wang, Xinyue Cai, Jianhua Han, Hang Xu, and Li Zhang. Reason2drive: Towards interpretable and chain-based reasoning for autonomous driving. In *European Conference on Computer Vision*, pp. 292–308. Springer, 2024.

NVIDIA, :, Yan Wang, Wenjie Luo, Junjie Bai, Yulong Cao, Tong Che, Ke Chen, Yuxiao Chen, Jenna Diamond, Yifan Ding, Wenhao Ding, Liang Feng, Greg Heinrich, Jack Huang, Peter Karkus, Boyi Li, Pinyi Li, Tsung-Yi Lin, Dongran Liu, Ming-Yu Liu, Langechuan Liu, Zhijian Liu, Jason Lu, Yunxiang Mao, Pavlo Molchanov, Lindsey Pavao, Zhenghao Peng, Mike Ranzinger, Ed Schmerling, Shida Shen, Yunfei Shi, Sarah Tariq, Ran Tian, Tilman Wekel, Xinshuo Weng, Tianjun Xiao, Eric Yang, Xiaodong Yang, Yurong You, Xiaohui Zeng, Wenyuan Zhang, Boris Ivanovic, and Marco Pavone. Alpamayo-r1: Bridging reasoning and action prediction for generalizable autonomous driving in the long tail, 2026. URL https://arxiv.org/abs/2511.00088.

Chenbin Pan, Burhaneddin Yaman, Tommaso Nesti, Abhirup Mallik, Alessandro G Allievi, Senem Velipasalar, and Liu Ren. Vlp: Vision language planning for autonomous driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 14760–14769, June 2024.

Hao Shao, Letian Wang, Ruobing Chen, Steven L Waslander, Hongsheng Li, and Yu Liu. Reasonnet: End-to-end driving with temporal and global reasoning. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 13723–13733. IEEE, 2023.

Hao Shao, Yuxuan Hu, Letian Wang, Guanglu Song, Steven L. Waslander, Yu Liu, and Hongsheng Li. Lmdrive: Closed-loop end-to-end driving with large language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 15120–15130, June 2024.

Chen Shi, Jinrui Xu, Shaoshuai Shi, Kehua Sheng, Bo Zhang, and Li Jiang. Drivewam: Video generative priors enable scalable world-action modeling for autonomous driving. *arXiv preprint arXiv:2605.28544*, 2026.

Shaoshuai Shi, Li Jiang, Dengxin Dai, and Bernt Schiele. Mtr++: Multi-agent motion prediction with symmetric scene modeling and guided intention querying. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(5):3955–3971, 2024.

Jeongwan Shin, Jaehyeon Kim, Donguk Ko, and Jaeho Choi. Can language models understand mmwave data? benchmarking large language models for mmwave radar-based human understanding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Findings*, pp. 2208–2219, June 2026.

Chonghao Sima, Katrin Renz, Kashyap Chitta, Li Chen, Hanxue Zhang, Chengen Xie, Jens Beißwenger, Ping Luo, Andreas Geiger, and Hongyang Li. Drivelm: Driving with graph visual question answering. In *European conference on computer vision*, pp. 256–274. Springer, 2024.

Chonghao Sima, Kashyap Chitta, Zhiding Yu, Shiyi Lan, Ping Luo, Andreas Geiger, Hongyang Li, and Jose M Alvarez. Centaur: Robust end-to-end autonomous driving with test-time training. *arXiv preprint arXiv:2503.11650*, 2025.

Wenchao Sun, Xuewu Lin, Yining Shi, Chuang Zhang, Haoran Wu, and Sifa Zheng. Sparsedrive: End-to-end autonomous driving via sparse scene representation. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 8795–8801. IEEE, 2025.

Xiaoyu Tian, Junru Gu, Bailin Li, Yicheng Liu, Yang Wang, Zhiyong Zhao, Kun Zhan, Peng Jia, Xianpeng Lang, and Hang Zhao. Drivevlm: The convergence of autonomous driving and large vision-language models. *arXiv preprint arXiv:2402.12289*, 2024.

Wenwen Tong, Chonghao Sima, Tai Wang, Li Chen, Silei Wu, Hanming Deng, Yi Gu, Lewei Lu, Ping Luo, Dahua Lin, et al. Scene as occupancy. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 8372–8381. IEEE, 2023.

Weijie Wei, Zhipeng Luo, Feng Ling, and Venice Erin Liong. Spatial-aware vision language model for autonomous driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 707–717, 2026.

Philipp Wolters, Johannes Gilg, Torben Teepe, and Gerhard Rigoll. Sparc-ad: A baseline for radar-camera fusion in end-to-end autonomous driving. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops*, pp. 1831–1841, October 2025.

Shuo Xing, Chengyuan Qian, Yuping Wang, Hongyuan Hua, Kexin Tian, Yang Zhou, and Zhengzhong Tu. Openemma: Open-source multimodal model for end-to-end autonomous driving. In *2025 IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)*, pp. 911–919. IEEE, 2025.

Yi Xu, Yuxin Hu, Zaiwei Zhang, Gregory P Meyer, Siva Karthik Mustikovela, Siddhartha Srinivasa, Eric M Wolff, and Xin Huang. Vlm-ad: End-to-end autonomous driving through vision-language model supervision. *arXiv preprint arXiv:2412.14446*, 2024a.

Zhenhua Xu, Yujia Zhang, Enze Xie, Zhen Zhao, Yong Guo, Kwan-Yee K Wong, Zhenguo Li, and Hengshuang Zhao. Drivegpt4: Interpretable end-to-end autonomous driving via large language model. *IEEE Robotics and Automation Letters*, 9(10):8186–8193, 2024b.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jing Zhou, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang, Le Yu, Lianghao Deng, Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 technical report, 2025. URL `https://arxiv.org/abs/2505.09388`.

Chenyu Yang, Yuntao Chen, Hao Tian, Chenxin Tao, Xizhou Zhu, Zhaoxiang Zhang, Gao Huang, Hongyang Li, Yu Qiao, Lewei Lu, et al. Bevformer v2: Adapting modern image backbones to bird's-eye-view recognition via perspective supervision. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 17830–17839. IEEE, 2023.

Yuqi Ye, Zijian Zhang, Junhong Lin, Shangkun Sun, Changhao Peng, and Wei Gao. AutoDrive-P^3: Unified chain of perception–prediction–planning thought via reinforcement fine-tuning. In *The Fourteenth International Conference on Learning Representations*, 2026. URL https://openreview.net/forum?id=CMU8GxwpUL.

Chengran Yuan, Zhanqi Zhang, Jiawei Sun, Shuo Sun, Zefan Huang, Christina Dao Wen Lee, Dongen Li, Yuhang Han, Anthony Wong, Keng Peng Tee, et al. Drama: An efficient end-to-end motion planner for autonomous driving with mamba. *arXiv preprint arXiv:2408.03601*, 2024.

Zhenlong Yuan, Chengxuan Qian, Jing Tang, Rui Chen, Zijian Song, Lei Sun, Xiangxiang Chu, Yujun Cai, Dapeng Zhang, and Shuo Li. Autodrive-r²: Incentivizing reasoning and self-reflection capacity for vla model in autonomous driving. In *International Conference on Learning Representations*, volume 2026, pp. 97934–97955, 2026.

Bozhou Zhang, Nan Song, and Li Zhang. Decoupling motion forecasting into directional intentions and dynamic states. *Advances in Neural Information Processing Systems*, 37:106582–106606, 2024.

Dapeng Zhang, Zhenlong Yuan, Zhangquan Chen, Chih-Ting Liao, Yinda Chen, Fei Shen, Qingguo Zhou, and Tat-Seng Chua. Reasoning-VLA: An efficient and spatial-guided general vision-language-action reasoning model for autonomous driving. In *Forty-third International Conference on Machine Learning*, 2026. URL https://openreview.net/forum?id=c4iSIrb6Iv.

Zongchuang Zhao, Xin Zhou, Tianyang Xu, Zhengyang Sun, Kaixuan Zhou, Honglin Li, Dingkang Liang, and Xiang Bai. Simwam: A simple world action model for end-to-end autonomous driving. *arXiv preprint arXiv:2608.07468*, 2026.

Wenzhao Zheng, Ruiqi Song, Xianda Guo, Chenming Zhang, and Long Chen. Genad: Generative end-to-end autonomous driving. In *European Conference on Computer Vision*, pp. 87–104. Springer, 2024a.

Wenzhao Zheng, Junjie Wu, Yao Zheng, Sicheng Zuo, Zixun Xie, Longchao Yang, Yong Pan, Zhihui Hao, Peng Jia, Xianpeng Lang, et al. Gaussianad: Gaussian-centric end-to-end autonomous driving. *arXiv preprint arXiv:2412.10371*, 2024b.

Yinan Zheng, Ruiming Liang, Kexin Zheng, Jinliang Zheng, Liyuan Mao, Jianxiong Li, Weihao Gu, Rui Ai, Shengbo Li, Xianyuan Zhan, et al. Diffusion-based planning for autonomous driving with flexible guidance. In *International conference on learning representations*, volume 2025, pp. 37207–37227, 2025.

Xingcheng Zhou, Xuyuan Han, Feng Yang, Yunpu Ma, Volker Tresp, and Alois Knoll. Opendrivevla: Towards end-to-end autonomous driving with large vision language action model. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pp. 13782–13790, 2026a.

Zewei Zhou, Tianhui Cai, Seth Zhao, Yun Zhang, Zhiyu Huang, Bolei Zhou, and Jiaqi Ma. Autovla: A vision-language-action model for end-to-end autonomous driving with adaptive reasoning and reinforcement fine-tuning. *Advances in Neural Information Processing Systems*, 38:27920–27956, 2026b.

A Related Work

End-to-End Autonomous Driving Methods. Traditional autonomous driving (AD) systems rely on modular pipelines comprising perception (Liao et al., 2022; Li et al., 2024b; Yang et al., 2023), prediction (Liu et al., 2021; Shi et al., 2024; Zhang et al., 2024), and planning (Cheng et al., 2024b;a), which often suffer from error accumulation and a lack of joint optimization across modules. To address this, end-to-end (E2E) AD methods (Shao et al., 2023; Jia et al., 2023; Yuan et al., 2024; Li et al., 2024a; Sima et al., 2025; Zheng et al., 2024b) have emerged to directly map sensor inputs to planning outputs, with early works like ST-P3 (Hu et al., 2022) pioneering this unified architecture. In particular, transformer-based architectures such as UniAD (Hu et al., 2023) and VAD (Jiang et al., 2023), OccNet (Tong et al., 2023), and SparseDrive (Sun et al., 2025), and recent generative approaches (Zheng et al., 2024a; Liao et al., 2025; Zheng et al., 2025; Shi et al., 2026; Zhao et al., 2026). Subsequently, generative approaches (Zheng et al., 2024a; Liao et al., 2025; Zheng et al., 2025) have been introduced to effectively model the complex future interactions between the ego vehicle and surrounding agents, as well as the inherent uncertainty of driving behaviors, thereby ensuring smooth trajectories. However, despite these advancements, pure E2E AD systems often exhibit limited adaptability in open-world, long-tail environments due to critical bottlenecks in explicit semantic reasoning (Zhou et al., 2026a;b).

B Dataset

This appendix describes the construction of dataset. We introduce the source data, define the seven tasks and their camera and radar inputs, describe how answers and rationales are constructed, and report the composition of the resulting dataset.

B.1 Training Data with Radar-Derived Physical Cues.

General SFT Data Training a radar-aware foundation model for autonomous driving requires unified supervision that jointly incorporates visual, radar, and language information. To this end, we leverage synchronized camera–radar driving clips and metadata from the NVIDIA PhysicalAI-Autonomous Vehicles dataset (NVIDIA et al., 2026), and transform them into multimodal instruction–response pairs using task-specific computational pipelines designed for each supervision target. The resulting training corpus spans seven categories of driving tasks, including object detection and tracking, ego and agent trajectory prediction, motion-state understanding, question answering, and scene description. These tasks are designed to exploit the complementary strengths of visual and radar observations, enabling ***RaPhy-AD*** to associate visual semantics with radar-derived 3D geometric and motion cues, such as object locations and radial velocities. Through this multi-task supervision, ***RaPhy-AD*** acquires radar-aware driving knowledge before being further optimized for structured reasoning and trajectory planning. Additional details on the training data construction are provided in Appendix B.

Structured Physical CoT Data. Beyond general instruction tuning, we construct structured physical CoT data to explicitly supervise ***RaPhy-AD*** on how radar-derived physical cues are incorporated into driving reasoning and decision-making. As illustrated in Fig. 2, the rationale supervision decomposes each driving sample into a sequence of structured reasoning components: the navigation command C, ego-motion state $\mathcal{R}_{\text{ego}}$, object-level physical states $\mathcal{R}_{\text{obj}}$, visual scene cues $\mathcal{R}_{\text{vis}}$, and driving-level reasoning $\mathcal{R}_{\text{drive}}$. These intermediate components are followed by the future ego trajectory Traj. Specifically, the ego-motion state is represented as $\mathcal{R}_{\text{ego}} = \{v_{\text{ego}}, \dot{\psi}_{\text{ego}}\}$, where v_{ego} and $\dot{\psi}_{\text{ego}}$ denote the ego speed and yaw rate. For each relevant object i, the physical state is represented as $\mathcal{R}_{\text{obj}}^{(i)} = \{r_i, \theta_i, n_i^{\text{radar}}, v_i^{\text{close}}\}$, where r_i and θ_i denote its range and azimuth, n_i^{radar} denotes the number of radar points associated with object i, and v_i^{close} denotes its closing velocity relative to the ego vehicle. The visual scene cues $\mathcal{R}_{\text{vis}}$ are designed to complement

radar-based physical reasoning by capturing scene information that cannot be directly inferred from ego-motion states or radar-derived object measurements. Specifically, it describes visually observable cues such as traffic signals, stop lines, crosswalks, the geometric structure of intersections and roundabouts, lane constraints, stationary obstacles, and environmental conditions. Based on these intermediate observations, the driving rationale $\mathcal{R}_{\text{drive}}$ provides an explicit, physically grounded explanation prior to trajectory generation, grounded in measurable quantities such as v_{ego}, $\dot{\psi}_{\text{ego}}$, r_i, θ_i, and v_i^{close}.

B.2 Data Collection

To construct the SFT dataset, we use driving data comprising camera images, radar measurements, ego-motion records, and annotations of surrounding objects. We describe the data source and selection criterion below.

Source Dataset. We use the NVIDIA PhysicalAI-Autonomous Vehicles dataset (NVIDIA et al., 2026), which contains approximately 1,700 hours of multimodal driving data organized into approximately 20 s clips. Each clip contains timestamped camera streams, point-level radar measurements, 3D object annotations, ego-motion, and sensor calibration. We use the front wide-angle camera, frontal radar measurements, and the corresponding annotations and metadata.

Clip Selection. The source dataset contains 177,891 clips. We exclude 17,130 clips without valid measurements from the frontal radar sensors, retaining 160,761 clips. We define seven tasks from the retained clips in the following subsection.

B.3 Task Definitions

The selected driving clips provide the source data for seven supervised fine-tuning tasks. We specify each task's objective, expected output, and camera, radar, and ego-motion inputs. For each task, we set the reference time t to the timestamp of the latest camera frame in its input and specify the timestamps of the other camera frames, radar scans, and ego-motion records relative to t.

Object detection. We identify and localize nearby road users at t. We use one front-camera frame at t, 20 radar scans at 0.05 s intervals over $[t - 0.95\,\text{s}, t]$, and one ego-motion record at $t - 1\,\text{s}$. For range–azimuth detection, the output is each road user's class, range, and azimuth. For 3D-box detection, the output is its class, position, dimensions, heading, planar velocity, and motion state.

Object tracking. Given object detections from preceding camera frames, we estimate current locations at t while preserving the assigned object identities. The input consists of five front-camera frames at 1 s intervals over $[t - 4\,\text{s}, t]$, 20 radar scans and 20 binned ego-motion entries at 0.25 s intervals over $[t - 4.75\,\text{s}, t]$, and object detections with assigned IDs from the four preceding camera frames. For each road user, the output is its ID, class, and current location, represented as a 2D image bounding box or range–azimuth coordinates.

Ego trajectory prediction. We predict the ego vehicle's motion over the 3 s following t. The inputs comprise two front-camera frames at $t - 1\,\text{s}$ and t, 20 radar scans and 20 binned ego-motion entries at 0.1 s intervals over $[t - 1.9\,\text{s}, t]$, and a high-level navigation command indicating `LEFT`, `RIGHT`, or `STRAIGHT`. The output specifies the ego vehicle's planar positions or its speed and signed yaw rate at $t + 1\,\text{s}$, $t + 2\,\text{s}$, and $t + 3\,\text{s}$.

Agent trajectory prediction. For a queried road user, we predict future locations at 1, 2, and 3 s after t. The input includes two front-camera frames at $t - 1\,\text{s}$ and t, together with 20 radar scans and 20 binned

ego-motion entries at 0.1 s intervals over $[t - 1.9\,\mathrm{s}, t]$. The instruction identifies the queried road user by its track ID, class, and current state (either as an image bounding box or range–azimuth coordinates). At each future time, the output gives the predicted location formatted as either a 2D image bounding box with coordinates normalized to $[0, 1000]$ or range–azimuth coordinates relative to the ego vehicle, or indicates that the road user has left the forward sector.

Motion-state understanding. We classify each road user in the forward sector as `moving` or `stationary` at t. We use two front-camera frames at $t - 1\,\mathrm{s}$ and t, 20 radar scans, and 20 binned ego-motion entries at 0.1 s intervals over $[t - 1.9\,\mathrm{s}, t]$. The output lists each road user's class, current location (represented as a 2D image bounding box or range–azimuth coordinates), and predicted motion state.

Question answering. Each question concerns a safety-critical driving situation and is answered by selecting one of five candidate options. The input contains 20 front-camera frames and 20 radar scans at 1 s intervals over $[t - 19\,\mathrm{s}, t]$, available binned ego-motion records, and the question and its answer options. We use the correct option letter as the answer target and include a natural-language rationale before the answer in rationale-supervised samples.

Scene description. The scene description task summarizes the ego vehicle and its surroundings at two temporal scales: the current scene at t and the full driving clip. For current-scene descriptions, we use front-camera frames and radar scans from the preceding 3 s and 20 binned ego-motion entries at 10 Hz ending at t. For clip summaries, we use 20 front-camera frames and 20 radar scans at 1 s intervals over $[t - 19\,\mathrm{s}, t]$ and the available ego-motion records.

B.4 Instruction–Response Pairs

We construct instruction–response pairs from the driving clips and annotations for the tasks defined in Sec. B.3. Each instruction contains a task-specific request, the corresponding camera frames and radar scans, sensor-availability information, and a textual representation of ego motion. Ego-motion records are represented by a temporal index and 32-bin indices for speed, acceleration, and yaw rate (e.g., `t0:s14a16y15`). Depending on the task, we also include previous detections, a queried road user's current state, a navigation command, or answer options. Responses contain either the answer alone or a rationale followed by the answer. For responses with a rationale, the instruction explicitly requests an explanation before the final answer. We construct these pairs programmatically, except for question answering, where an LLM generates the questions, answer choices, answers, and rationales.

Object detection. We construct detection answers from the annotated 3D boxes at t, listing up to eight road users within 40 m ahead in order of increasing distance. For range–azimuth detection, we calculate planar range and signed azimuth from each box center and pair these values with the object class. For 3D-box detection, we retain the annotated center and dimensions, quantize heading relative to the ego-forward axis into 12 bins at 30° intervals, and calculate planar velocity when at least three track observations are available. We assign motion state using a planar-speed threshold of $1\,\mathrm{m/s}$. Each instruction specifies the selection and ordering criteria and requests the attributes of the corresponding variant. The answer lists the selected objects with these attributes in the requested order. For the rationale, we project the annotated boxes into the camera view and associate radar returns with each box through a 3D point-in-box test. We express the resulting evidence as an object-wise account of camera-view azimuth and the count, median range, and median azimuth of associated radar returns, explicitly noting objects without associated returns. These measurements provide the sensor evidence described in the rationale, whereas target coordinates are derived from the annotated box centers. Table 7 illustrates both variants.

Object tracking. We assign consecutive IDs to annotated tracks in order of first appearance within each clip and preserve these IDs across observations. For each sample, annotations from the four preceding camera frames are converted into lists of IDs, classes, and locations, indexed by their time relative to t. We insert these lists into the instruction as `Previous detections`, followed by a request to update the object list at t, reuse existing IDs, and assign new IDs to newly observed objects. Current annotations form the answer, with each ID paired with its class and either range–azimuth coordinates or a 2D bounding box normalized to $[0, 1000]$. The rationale relates each object's previous observation to its current camera and radar evidence and identifies newly observed objects. Occlusion is estimated from overlap with nearer image boxes and is reported when the overlap exceeds 50%. Associated radar return counts, ranges, and azimuths are included where available. Table 8 shows how the historical lists in the instruction correspond to the updated IDs and locations in the response.

Ego trajectory prediction. The source dataset does not provide high-level navigation commands. Following prior work (Hu et al., 2023; Jiang et al., 2023), we derive them from future ego trajectories. We classify the ego yaw change over the subsequent 3 s as `LEFT`, `RIGHT`, or `STRAIGHT` using a $\pm 15^\circ$ threshold. We include the resulting command in the instruction together with a request for either planar position offsets or speed and signed yaw rate. The corresponding targets are obtained from future ego-motion records and listed chronologically under `+1s`, `+2s`, and `+3s`. The navigation command is a high-level route label constructed offline from the ground-truth future trajectory. At inference, only the resulting categorical command token is included in the input instruction as route conditioning; the underlying future yaw values and ego-motion records are not provided to the model, nor are any future camera or radar observations. Sensor inputs remain restricted to observations available at or before the reference time t. This protocol therefore evaluates trajectory prediction conditioned on an oracle route command, rather than requiring the model to infer the route from past observations alone; the command conveys high-level intent derived from the ground-truth future motion. To construct the rationale, we summarize the future motion represented by these targets in terms of longitudinal and steering behavior. We categorize speed changes using a $\pm 1.5\,\mathrm{m/s}$ threshold as `accelerate`, `brake`, or `hold speed`, and yaw changes using a $\pm 8^\circ$ threshold as `turn left`, `turn right`, or `go straight`. We combine these descriptions with the current ego speed to form a rationale that precedes the numerical trajectory, as shown in Table 9. These target-derived rationale labels are constructed offline and used only as supervised response targets, not as input instructions. At inference, the model generates the rationale and numerical trajectory autoregressively from the available observations and command; neither the ground-truth rationale nor the ground-truth trajectory is supplied as an input.

Agent trajectory prediction. We select target agents with at least 2 m of planar displacement over the observed history. Each instruction identifies the selected agent by its track ID, class, and current location, then asks for its locations over the next 3 s in the specified coordinate representation. We follow the same annotated track to obtain the future targets and express them as normalized image bounding boxes or range–azimuth pairs at 1, 2, and 3 s. The response lists these targets by prediction horizon and explicitly indicates when the agent leaves the forward sector. The rationale combines the agent's current location with associated radar measurements. Using the median radial velocity and thresholds of $\pm 0.5\,\mathrm{m/s}$, we describe the agent as `closing`, `receding`, or `holding its range`. This description provides motion evidence before the future locations in Table 10.

Motion-state understanding. We differentiate each object's world-frame position history to obtain planar velocity and assign motion-state labels using a speed threshold of $1\,\mathrm{m/s}$. The instruction asks which road users within 40 m ahead are moving or stationary and specifies whether they should be identified by image bounding boxes or range–azimuth coordinates. We group the annotated objects by their assigned labels and construct the answer as `moving:` and `stationary:` lists, each containing object classes and locations. For the rationale, we compare observed radar radial velocity and camera azimuth with their expected values under a stationary-object assumption. We report the expected and observed values and their residuals, using

thresholds of $1\,\mathrm{m/s}$ and 1°, respectively, to describe evidence of motion. The response presents the sensor comparisons before the grouped labels, as illustrated in Table 11.

Question answering. For the question answering task, we use Gemini 2.5 Flash. The model is given 20 camera frames uniformly sampled from each clip, together with structured textual summaries of the corresponding object annotations, radar statistics, and ego motion. We prompt the model to identify a safety-critical moment and generate questions across perception, anticipation, causal reasoning, and planning. Each generated item contains a question, five candidate answers, the correct option, and a rationale based on the supplied evidence. We place the generated question and options, labeled A–E, in the instruction and use the correct option letter as the answer. The first frame explicitly mentioned in the question determines its temporal reference; otherwise, the final frame is used. For rationale-supervised pairs, the instruction additionally requests step-by-step reasoning, and the generated rationale is placed before the option letter. Answer-only pairs retain the option letter without a separate rationale. Table 12 shows a completed pair.

Scene description. We generate scene descriptions using rule-based language templates instantiated with precomputed attributes of ego motion, road users, and radar observations. Current-scene templates express attributes at the reference time, such as ego speed, acceleration, and turning behavior. Clip-summary templates additionally describe aggregate properties, including the distance traveled and the range of ego speeds over the clip. Template variants are selected deterministically from clip and frame identifiers. We pair the resulting text with an instruction to describe the scene and use the description directly as the response, without a separate rationale. Table 13 provides examples of the current-scene and clip-summary variants. Tables 7–13 show example instruction–response pairs from the constructed SFT dataset for each task. The tables display the textual components of the training samples; front-camera video frames and radar point cloud tensors provided to the model during training are omitted for readability. Intermediate ego-motion bins and repeated object entries are abbreviated by ellipses, and repeated instruction wording is shortened in the examples; the full samples are used during training.

Table 7: Example instruction–response pairs for Object Detection.

Task / Variant	Instruction–response pair
Object Detection (Range–Azimuth)	**Instruction.** Sensors present: camera, ego motion, imaging long-range radar. Ego motion (binned, at t=12s): t0:s17a13y16 At frame 13. List the road users within 40m ahead, nearest first, up to 8, with class, range and azimuth. Reason step by step first, then give the answer. **Response.** `{"rationale": "heavy_truck: camera az +0deg, radar 20 returns at 34m az +1deg. Range comes from the radar where it has a return and from the camera alone where it does not.", "answer": "heavy_truck 36m az +1deg"}`
Object Detection (3D Box)	**Instruction.** Sensors present: camera, ego motion, short-range radar. Ego motion (binned, at t=15s): t0:s2a25y15 At frame 16. List the road users within 40m ahead, nearest first, up to 8, as a 3D box: class, centre (x, y, z) in metres with x forward, y left and z up, then size as length x width x height in metres, heading sector (0–11), planar velocity (vx, vy) in m/s, and motion state (moving or stopped). Reason step by step first, then give the answer. **Response.** `{"rationale": "automobile: camera az +16deg, radar 13 returns at 18m az +11deg; automobile: camera az -53deg, radar 1 return at 16m az -47deg; ...; automobile: camera az +12deg, no radar return. Range comes from the radar where it has a return and from the camera alone where it does not.", "answer": "automobile (15.7m, 3.7m, 0.7m) size 4.5mx1.9mx1.6m heading 6 vel +1.2m/s +0.1m/s moving; automobile (11.5m, -13.8m, 1.0m) size 4.5mx1.9mx1.5m heading 6 vel +0.0m/s +0.0m/s stopped; ...; automobile (35.1m, 7.0m, 0.1m) size 4.5mx1.9mx1.5m heading 6 vel +1.7m/s +0.1m/s moving"}`

Table 8: Example instruction–response pairs for Object Tracking.

Task / Variant	Instruction–response pair
Object Tracking (Range–Azimuth)	**Instruction.** Sensors present: camera, ego motion, imaging long-range radar. Ego motion (binned, 5 s at 4 Hz, ending at t=19s): t0:s19a16y15 . . . t19:s19a17y16 At frame 20. Previous detections: t-4s: none; . . . ; t-2s: none t-1s: #26 automobile 180 m az +1 deg Continue tracking: report each current road user's ID, class, range, and azimuth, preserving existing IDs and assigning new ones as needed. Reason step by step first, then give the answer. **Response.** `{"rationale": "#26 automobile: was at 180m az +1deg, camera az +2deg, no radar return. Range comes from the radar where it has a return and from the camera alone where it does not.", "answer": "#26 automobile 134m az +1deg"}`
Object Tracking (2D Box)	**Instruction.** Sensors present: camera, ego motion, imaging long-range radar. Ego motion (binned, 5 s at 4 Hz, ending at t=16s): t0:s1a13y10 . . . t19:s0a18y12 At frame 17. Previous detections: t-4s: #1 automobile [416, 665, 598, 804]; . . . ; #11 automobile [613, 663, 717, 741] . . . t-1s: #1 automobile [412, 659, 589, 793]; . . . ; #11 automobile [608, 666, 710, 743] Continue tracking: report each current road user's class, ID, and image box [x1, y1, x2, y2] normalised to 0–1000, preserving existing IDs and assigning new ones as needed. Reason step by step first, then give the answer. **Response.** `{"rationale": "#20 automobile: was at 7m az +49deg, no radar return; #1 automobile: was at 13m az +0deg, camera az +3deg, radar 7 returns at 13m az +5deg; ...; #11 automobile: was at 21m az −17deg, camera az −15deg, 100% occluded, radar 5 returns at 21m az −18deg. Range comes from the radar where it has a return and from the camera alone where it does not.", "answer": "#1 automobile [383, 662, 564, 800]; #10 automobile [502, 672, 663, 784]; ...; #11 automobile [582, 669, 685, 747]"}`

Table 9: Example instruction–response pairs for Ego Trajectory Prediction.

Task / Variant	Instruction–response pair
Ego Trajectory (xy offsets)	**Instruction.** Sensors present: camera, ego motion, short-range radar. Ego motion (binned, 2 s at 10 Hz, ending at t=9s): t0:s1a15y15 t1:s1a15y15 . . . t10:s2a27y15 . . . t18:s4a20y15 t19:s4a20y14 At frame 10. Navigation command: STRAIGHT. Predict the ego vehicle's path over the next 3 seconds as (x, y) offsets in metres. Reason step by step first, then give the answer. **Response.** `{"rationale": "The ego vehicle is travelling at 4.6m/s and will brake and go straight. At that speed it covers about 5m per second.", "answer": "+1s (+4.8m, −0.1m); +2s (+9.5m, −0.1m); +3s (+13.2m, −0.2m)"}`
Ego Trajectory (Speed / Yaw)	**Instruction.** Sensors present: camera, ego motion, imaging long-range radar. Ego motion (binned, 2 s at 10 Hz, ending at t=15s): t0:s31a13y15 t1:s31a16y15 . . . t10:s31a14y15 . . . t18:s31a16y15 t19:s31a12y15 At frame 16. Navigation command: STRAIGHT. Predict the ego vehicle's speed and yaw rate over the next 3 seconds. Reason step by step first, then give the answer. **Response.** `{"rationale": "The ego vehicle is travelling at 34.5m/s and will hold speed and go straight. At that speed it covers about 35m per second.", "answer": "+1s 34.7m/s, yaw −0.9deg/s; +2s 34.7m/s, yaw −0.6deg/s; +3s 34.6m/s, yaw −0.6deg/s"}`

Table 10: Example instruction–response pairs for Agent Trajectory Prediction.

Task / Variant	Instruction–response pair
Agent Trajectory (Bounding Box)	**Instruction.** Sensors present: camera, ego motion, imaging long-range radar. Ego motion (binned, 2 s at 10 Hz, ending at t=13s): t0:s14a12y17 t1:s14a11y17 . . . t10:s13a12y18 . . . t18:s13a14y18 t19:s13a15y17 At frame 14. Track #14 is a heavy_truck at [436, 661, 448, 688]. Where will it be over the next 3 seconds? Answer as an image bounding box [x1, y1, x2, y2] normalised to 0-1000, or say 'leaves the forward sector' for a horizon it does not reach. Reason step by step first, then give the answer. **Response.** `{"rationale": "#14 heavy_truck is at 121m az +7deg now, camera az +7deg, radar 19 returns at 119m, radial +6.1m/s, so it is receding: radial velocity times time is the change in range.", "answer": "+1s [421, 665, 433, 692]; +2s [406, 670, 420, 696]; +3s [398, 668, 414, 693]"}`
Agent Trajectory (Range–Azimuth)	**Instruction.** Sensors present: camera, ego motion, imaging long-range radar. Ego motion (binned, 2 s at 10 Hz, ending at t=17s): t0:s1a13y16 t1:s1a14y15 . . . t10:s1a12y16 . . . t18:s0a13y16 t19:s0a13y15 At frame 18. Track #577 is a bus at 8m, azimuth -25deg. Where will it be over the next 3 seconds? Answer as range in metres and azimuth in degrees, or say 'leaves the forward sector' for a horizon it does not reach. Reason step by step first, then give the answer. **Response.** `{"rationale": "#577 bus is at 8m az −25deg now, camera az −17deg, radar 16 returns at 9m, radial −0.6m/s, so it is closing: radial velocity times time is the change in range.", "answer": "+1s 7m az −26deg; +2s 7m az −27deg; +3s 7m az −27deg"}`

Table 11: Example instruction–response pairs for Motion-State Understanding.

Task / Variant	Instruction–response pair
Motion State (Bounding Box)	**Instruction.** Sensors present: camera, ego motion, short-range radar. Ego motion (binned, 2 s at 10 Hz, ending at t=6s): t0:s0a14y15 t1:s0a13y15 . . . t10:s0a14y15 . . . t18:s0a15y15 t19:s0a15y16 At frame 7. Which of the road users within 40m ahead are moving and which are stationary? Use the radar Doppler and the camera. Identify each as an image bounding box [x1, y1, x2, y2] normalised to 0-1000. Reason step by step first, then give the answer. **Response.** `{"rationale": "The ego is travelling at 0.3m/s, yaw +0.0deg/s. automobile at 7m az +31deg: radar expects −0.2m/s if stationary, measures +1.4 (residual 1.7), camera expects az +31deg if stationary, sees +31 (residual 0.1deg); ...; automobile at 10m az −0deg: radar expects −0.3m/s if stationary, measures −0.3 (residual 0.0), camera expects az −0deg if stationary, sees −0 (residual 0.0deg); ...; A radial residual above 1m/s or a bearing residual above 1deg means the object is moving; the Doppler is blind to an object crossing the ego's path and the bearing is blind to one approaching head-on, so either is enough.", "answer": "moving: automobile [653, 462, 884, 701], automobile [465, 471, 521, 553]. stationary: automobile [0, 420, 302, 820], ..., automobile [483, 465, 525, 533]."}`

Table 12: Example instruction–response pairs for Question Answering.

Task / Variant	Instruction–response pair
Question Answering (Multiple choice)	**Instruction.** Sensors present: camera, ego motion, imaging long-range radar. Ego motion (binned, t0..t13): t0:s4a14y16 t1:s4a15y15 . . . t7:s9a17y16 . . . t12:s13a19y16 t13:s14a18y16 At frame 14, should the ego attempt to overtake the dark heavy truck directly ahead? A. Yes, it is safe. B. No, because the ego vehicle is preparing to turn right. C. No, due to road conditions and oncoming traffic. D. Yes, if accelerating faster. E. Yes, if the truck slows down. Reason step by step, then answer with the letter. **Response.** `{"rationale": "At Frame 14, the ego vehicle is in a construction zone, indicated by the barriers on the left. This limits lateral maneuverability and creates a confined driving environment. Additionally, there are large oncoming vehicles (a white heavy truck with 'NUSS' on its side and a white trailer behind it) in the adjacent lane. Attempting to overtake in these road conditions, which include limited space and opposing traffic, would be unsafe. Therefore, the ego vehicle should not attempt to overtake due to road conditions and oncoming traffic.", "answer": "C"}`

Table 13: Example instruction–response pairs for Scene Description.

Task / Variant	Instruction–response pair
Scene Description (Current scene)	**Instruction.** Sensors present: camera, ego motion, short-range radar. Ego motion (binned, 2 s at 10 Hz, ending at t=9s): t0:s1a15y15 t1:s1a15y15 t2:s1a16y15 t3:s1a16y15 t4:s1a16y15 t5:s1a18y15 t6:s1a20y15 t7:s1a21y15 t8:s2a22y15 t9:s2a24y15 t10:s2a27y15 t11:s2a28y15 t12:s3a28y15 t13:s3a28y15 t14:s3a26y15 t15:s3a26y15 t16:s4a25y15 t17:s4a22y15 t18:s4a20y15 t19:s4a20y14 Describe the scene. **Response.** `The ego vehicle is travelling at 15.2 m/s, decelerating at -0.4 m/s`^{2} `while turning left at 2.1 deg/s.`
Scene Description (Clip summary)	**Instruction.** Sensors present: camera, ego motion, imaging long-range radar. Ego motion (binned, t0..t12): t0:s9a24y16 t1:s12a24y16 t2:s12a16y15 t3:s12a16y16 t4:s12a13y18 t5:s12a16y19 t6:s12a17y16 t7:s12a16y15 t8:s12a13y15 t9:s11a12y16 t10:s10a10y16 t11:s9a10y15 t12:s7a6y16 Describe the scene. **Response.** `Over 20 s, the ego vehicle covers 285 m, with speed between 12.1 and 18.4 m/s. It accelerates from 12.1 m/s to 18.4 m/s over the first 10 s, then maintains speed while gently curving right. There are 14 vehicles ahead, mostly moving smoothly in their lanes.`

B.5 Dataset Statistics

From the instruction–response pairs generated in the preceding sections, we programmatically remove invalid samples and exact duplicates. We reserve 1,210 clips, corresponding to 1,287 evaluation anchors, as the test set, while using all remaining samples for training. The test set is curated to cover diverse driving scenarios in which accurate physical information is particularly important for decision-making, and includes 644 radar-equipped clips adopted from the evaluation split used in prior work on the NVIDIA PhysicalAI-Autonomous Vehicles dataset. Supervised fine-tuning is performed with weighted sampling over tasks from this training pool. Table 14 summarizes the $2{,}984{,}608$ instruction–response pairs consumed across the two supervised runs by task and response format.

Of these pairs, $1{,}498{,}652$ contain answer-only responses and $1{,}485{,}956$ contain rationale-augmented responses. Object detection and ego trajectory prediction account for $1{,}031{,}296$ and $914{,}458$ pairs, respectively, together representing 65.2% of the total. Question answering appears in both response formats, while

Table 14: Statistics of samples used for Training.

Task	w/o rationale	w/ rationale	Total
Object detection	515,648	515,648	1,031,296
Ego trajectory prediction	457,229	457,229	914,458
Agent trajectory prediction	175,258	175,258	350,516
Motion-state understanding	155,785	155,785	311,570
Object tracking	77,893	77,893	155,786
Question answering	77,893	77,893	155,786
Scene description	38,946	–	38,946
Structured physical CoT	–	26,250	26,250
Total	**1,498,652**	**1,485,956**	**2,984,608**

scene description uses only answer-only responses. The structured physical CoT data contribute 26,250 rationale-augmented pairs with the output schema `visual_cue`, `object`, `rationale`, and `answer`.

C Supervised Fine-Tuning Details

C.1 Stage and Initialisation

We initialise the vision–language backbone from the pretrained `Qwen3-VL-8B-Instruct` checkpoint released by the Qwen team. The Polar-Time Radar Encoder starts from random initialisation and is trained jointly with the language model and radar connector during supervised fine-tuning.

Table 15 summarizes the optimization settings used for supervised fine-tuning. We perform full-parameter fine-tuning without adapters using AdamW with $\beta = (0.9, 0.95)$ and a weight decay of 0.01. The learning rate is scheduled with OneCycle, reaching a peak of 1×10^{-4} with a 3% warm-up period, and gradients are clipped at 5.0. Training is conducted in `bf16` using FSDP2 full sharding across seven devices, with gradient accumulation of 5 and an effective batch size of 350. The maximum sequence length is set to $4{,}352$ tokens.

Table 15: Optimisation settings for supervised fine-tuning.

Setting	Value
Stage	`full` (no adapters)
Optimiser	AdamW, $\beta = (0.9,\ 0.95)$, weight decay 0.01
Peak learning rate	1×10^{-4}
Schedule	OneCycle, warm-up $= 3\%$ of total steps
Gradient clipping	5.0
Master dtype	`bf16`
Devices	7 (FSDP2, full-shard)
Effective batch	$10 \times 5 \times 7 = 350$
Maximum sequence length	$4{,}352$ tokens

C.2 OBJECTIVE

The loss is the causal language-modelling cross-entropy $\mathcal{L}_{\text{CE}}$ plus four auxiliary terms:

$$\mathcal{L} \;=\; \mathcal{L}_{\text{CE}} \;+\; \lambda_{\text{ego}}\,\mathcal{L}_{\text{ego}} \;+\; w(t)[\lambda_{\text{num}}\,\mathcal{L}_{\text{num}} + \lambda_{\text{unit}}\,\mathcal{L}_{\text{unit}} + \lambda_{\text{gate}}\,\mathcal{L}_{\text{gate}}] \;+\; \lambda_{\text{router}}\,\mathcal{L}_{\text{router}}, \quad (8)$$

with the ramp $w(t) = \min(1,\ t/T_{\text{warm}})$ and $T_{\text{warm}} = 1{,}500$ steps.

- $\mathcal{L}_{\text{ego}}$ is cross-entropy restricted to the ego-state span. That span is about 20 tokens while the object list runs 300–400, so without the extra weight the ego field is drowned in the token average; an earlier run collapsed to three distinct speed values out of 405 possible.
- $\mathcal{L}_{\text{num}}$ penalises the *distance* between the predicted and target number token rather than treating them as unrelated classes, so that predicting 37 m for 38 m costs less than predicting 3 m. It is ramped in because, before the answer has the right shape, it pulls every field toward its marginal mean.
- $\mathcal{L}_{\text{unit}}$ penalises confusion between number tokens that share a value but differ in unit; pairs that differ only in a metre unit are weighted $3\times$.
- $\mathcal{L}_{\text{gate}}$ is an additional gate on the unit field of the emitted number token.
- $\mathcal{L}_{\text{router}}$ balances the radar encoder's routed experts; it is identically zero for the dense encoder used here.

Table 16: Loss weights.

Term	Symbol	Weight
Language-modelling CE	$\mathcal{L}_{\text{CE}}$	1.0
Ego-state CE	λ_{ego}	0.3
Number distance	λ_{num}	0.1
Unit confusion	λ_{unit}	0.5
metre-pair multiplier		3.0
Unit gate	λ_{gate}	0.3
Router balance	λ_{router}	0.01
Number-loss warm-up	T_{warm}	1,500 steps

C.3 VOCABULARY EXTENSION

We extend the base tokenizer from 151,669 to 163,571 entries, adding 11,902 new token IDs in total (Table 17). The number-literal list contains 5,080 entries, of which 10 already exist in the base vocabulary; it therefore contributes 5,070 new IDs. The trajectory block adds 3,000 trajectory tokens and one `<|traj_pad|>` token. Marker tokens and the + token reuse existing IDs and do not increase the vocabulary size.

C.4 NUMBER-TOKEN GEOMETRY

The 5,080 number literals are initialised so that their embeddings encode value (a sinusoidal code of the literal's numeric value) and unit (a rotation of 30° per unit). Because full fine-tuning is free to destroy that geometry, it is restored every 200 optimiser steps by `relax_value_geometry`, which pulls each literal's embedding back toward its value-consistent position with weight 1.0. The restoration edits the embedding table directly, so it is applied only on ranks that hold the full table.

Table 17: Vocabulary extension relative to the base Qwen3-VL-8B-Instruct tokenizer. Counts indicate newly added token IDs.

Token group	New IDs	Details
Radar placeholders	3	Radar input markers
Number literals	5,070	5,080 listed literals, including 10 existing entries
Trajectory tokens	3,001	3,000 trajectory tokens and one padding token
Additional unit literals	3,828	Newly added unit literals
Marker and + tokens	0	Already present in the base vocabulary
Total	**11,902**	151,669 → 163,571

D IMPLEMENTATION

Backbone and Radar Encoder Configuration. We use `Qwen3-VL-8B-Instruct` as the pretrained vision–language backbone. The radar encoder is randomly initialised and jointly optimised during SFT, as described in Appendix C. Table 18 summarises the Polar-Time Radar Encoder configuration. The polar grid has $N_\theta = 16$ azimuth cells spanning $[-60°, 60°]$ and $N_r = 4$ range rings spanning 0–40 m, corresponding to $7.5°$ and 10 m per cell, respectively. Each frame has $N_q = N_\theta N_r = 64$ polar queries; the learnable `polar_query` parameter has shape 64×384. With $T_\mathrm{r} = 10$ temporal slots, the encoder produces $T_\mathrm{r} N_q = 640$ radar tokens. Its nine blocks comprise four spatial Transformer blocks, three temporal Transformer blocks, and two pooling blocks (`frame_pool` and `polar_pool_block`).

Table 18: Configuration of the Polar-Time Radar Encoder.

Setting	Value
Azimuth cells, N_θ	16 over $[-60°, 60°]$ ($7.5°$ each)
Range rings, N_r	4 over 0–40 m (10 m each)
Polar queries per frame, N_q	$64 = N_\theta N_r$
Temporal slots, T_r	10
Output radar tokens	$640 = T_\mathrm{r} N_q$
Embedding width, D	384
Attention heads	8
Spatial Transformer blocks	4
Temporal Transformer blocks	3
Pooling blocks	2
Total blocks	9
Frame queries	10
Encoder parameters	19.6 million

Point Embedding Implementation Details. We use $K = 8$ Fourier frequency bands, yielding 16 Fourier features per geometric coordinate. Concatenating the Fourier features of the three coordinates with the original 8-dimensional point representation produces a 56-dimensional input, which the point-wise MLP maps to an embedding of dimension $D = 384$.

E RLVR OPTIMIZATION DETAILS

For each multimodal prompt q, the behavior policy $\pi_{\theta_\mathrm{old}}$ samples a group of G completions $\{o^{(i)}\}_{i=1}^{G}$. Each completion receives a verifiable reward r_i, which is normalized within the group to obtain the relative

advantage:

$$\mu_q = \frac{1}{G}\sum_{j=1}^{G} r_j, \qquad \sigma_q = \sqrt{\frac{1}{G}\sum_{j=1}^{G}(r_j - \mu_q)^2}, \qquad A_i = \frac{r_i - \mu_q}{\sigma_q}. \tag{9}$$

When the reward variance within a group is negligible, we set $A_i = 0$. This group-relative normalization removes the need for a separate value network.

To account for the large variation in response length across tasks, we compute the policy ratio using the length-normalized log-likelihood of each completion. Let

$$\bar{\ell}_\theta(o^{(i)} \mid q) = \frac{1}{|o^{(i)}|}\sum_{t=1}^{|o^{(i)}|} \log \pi_\theta\left(o_t^{(i)} \mid q, o_{<t}^{(i)}\right). \tag{10}$$

The corresponding importance ratio is

$$\rho_i(\theta) = \exp\left(\bar{\ell}_\theta(o^{(i)} \mid q) - \bar{\ell}_{\theta_{\text{old}}}(o^{(i)} \mid q)\right). \tag{11}$$

We then optimize the clipped GRPO objective

$$\mathcal{L}_{\text{GRPO}} = -\frac{1}{B}\sum_{i=1}^{B} \min\left(\rho_i(\theta)A_i,\ \text{clip}\left(\rho_i(\theta), 1-\epsilon, 1+\epsilon\right) A_i\right). \tag{12}$$

Unlike GRPO formulations that employ explicit KL regularization, our final configuration uses neither a frozen reference policy nor a KL penalty. Instead, the clipped importance ratio with respect to the behavior policy constrains the magnitude of the policy update.

Reward Implementation Details For object-centric tasks, predicted and ground-truth objects are first matched according to their spatial proximity. The object reward combines matching quality with the correctness of physical attributes:

$$R_{\text{obj}} = \frac{1}{2}\text{F1}_{\text{match}} + \frac{1}{2}\frac{1}{|\mathcal{M}|}\sum_{(i,j)\in\mathcal{M}} S_{\text{attr}}(\hat{o}_i, o_j), \tag{13}$$

where $\mathcal{M}$ denotes the set of matched objects and S_{attr} evaluates attributes such as object class, motion state, and spatial quantities. For ego-trajectory prediction, the reward accounts for both prediction coverage and geometric error:

$$R_{\text{traj}} = C_{\text{hor}}\frac{1}{T'}\sum_{t=1}^{T'} 2^{-\|\hat{\mathbf{p}}_t - \mathbf{p}_t\|_2/\delta_{\text{traj}}}, \tag{14}$$

where $C_{\text{hor}} = T'/T$ measures the fraction of the requested horizon that is successfully generated and δ_{traj} controls the distance tolerance. Similarly, agent-trajectory rewards combine distance and bearing errors:

$$R_{\text{agent}} = \frac{1}{2T'}\sum_{t=1}^{T'}\left(2^{-d_t/5\text{m}} + 2^{-|\Delta\theta_t|/10^\circ}\right). \tag{15}$$

For numerical reasoning tasks, we extract all predicted physical quantities and compare them with their corresponding targets using relative numerical error. For a quantity pair $(\hat{v}_j, v_j)$, we compute

$$e_j^{\text{rel}} = \frac{|\hat{v}_j - v_j|}{|v_j| + \epsilon}, \tag{16}$$

and aggregate the resulting numerical agreement scores over all verifiable quantities. Question-answering tasks use exact-choice accuracy, while scene-description rewards combine numerical consistency and claim-level F1, with an additional penalty for explicit contradictions.

For structured physical reasoning, the output is evaluated by multiple specialized verifiers rather than by a single scalar metric. The composite reward used in our final configuration is

$$\begin{aligned} R_{\text{reason}} = {} & 0.46\, V_{\text{traj}} + 0.20\, V_{\text{end}} + 0.10\, V_{\text{risk}} + 0.05\, V_{\text{obj}} + 0.03\, V_{\text{unit}} \\ & + 0.03\, V_{\text{ans-unit}} + 0.05\, V_{\text{vis}} + 0.05\, V_{\text{ego}} + 0.02\, V_{\text{rat}} + 0.01\, V_{\text{box3d}}. \end{aligned} \tag{17}$$

Here, V_{traj} evaluates the predicted future trajectory, V_{end} separately verifies the final trajectory endpoint, and V_{risk} checks whether the model correctly identifies safety-critical approaching objects. V_{obj} verifies object-level physical quantities through spatial matching, while V_{unit} and $V_{\text{ans-unit}}$ check the physical units associated with object attributes and trajectory coordinates, respectively. V_{vis} evaluates directly observable scene cues such as traffic signals, stop lines, and intersection structure, and V_{ego} verifies ego-motion states and the driving command. Importantly, V_{rat} does not score the linguistic style of the rationale. Instead, it extracts the physical quantities explicitly referenced in the reasoning and verifies whether they agree with the corresponding values derived from the ground-truth annotations.

Reward Function. Our RLVR framework uses a collection of deterministic verifiers that explicitly compare generated outputs with ground-truth annotations and physical quantities derived from them. Unlike preference-based rewards, each verifier evaluates a measurable aspect of the output, including trajectory accuracy, safety-critical objects, radar-derived physical quantities, ego motion, visual cues, physical units, and quantities cited in the rationale. For a valid structured response, the overall reward is computed as

$$R_{\text{reason}} = \sum_j w_j s_j, \qquad \sum_j w_j = 1, \tag{18}$$

where s_j denotes the score produced by verifier j.

For distance-based verification, we use $d(e; h) = h/(h+e)$, which yields a score of 1 at zero error and 0.5 at the tolerance h. Predicted and ground-truth objects are greedily matched one-to-one in the bird's-eye-view plane within $2\,\text{m}$. For trajectory verification, we use an adaptive tolerance $\eta_h = \max(0.05\ell_h, 1\,\text{m})$, where ℓ_h denotes the ground-truth travel distance at horizon h.

Because the structured reasoning output follows a six-field JSON schema, malformed responses are assigned a graded format reward:

$$R_{\text{fmt}} = \begin{cases} -1 + 0.4\,\dfrac{k}{6}, & \text{if the output cannot be parsed,} \\ -0.5\left(1 - \dfrac{m}{6}\right), & \text{if it is parsed but contains fewer than six fields,} \end{cases} \tag{19}$$

where k is the number of recoverable field names and m is the number of valid fields present. A graded penalty is used instead of a constant negative reward to preserve reward variation within a GRPO group, since identical rewards would yield zero group-relative advantage.

For the rationale verifier, let $\hat{\mathcal{C}}$ and $\mathcal{C}^*$ denote the sets of unit-bearing quantities extracted from the generated and reference rationales, respectively, and let $\mathcal{V}$ denote the set of verifiable physical quantities derived from the scene annotations. Rather than evaluating linguistic similarity, s_{rat} measures whether the quantities explicitly cited in the generated reasoning agree with these ground-truth physical values.

Table 19: Verifiable reward components for structured physical reasoning. Each component is computed programmatically from the generated output and ground-truth annotations.

Verifier	Computation	Weight	Tolerance
Trajectory	$s_{\text{ans}} = \frac{\lvert\mathcal{H}\rvert}{\lvert\mathcal{H}^*\rvert}\frac{\lvert\mathcal{H}\rvert}{\lvert\hat{\mathcal{H}}\rvert} d(\bar{e}; \eta_{h_{\max}})$, where $\bar{e} = \frac{1}{\lvert\mathcal{H}\rvert}\sum_{h\in\mathcal{H}} \lVert\hat{\mathbf{p}}_h - \mathbf{p}_h\rVert_2$.	0.46	Adaptive (η_h)
Final horizon	$s_{\text{end}} = d(\lVert\hat{\mathbf{p}}_{h_{\max}} - \mathbf{p}_{h_{\max}}\rVert_2; \eta_{h_{\max}})$; 0 if the final horizon is missing.	0.20	Adaptive (η_h)
Risk	F1 over matched safety-critical objects satisfying $g(o) \leq 2\,\text{s}$ and $v^{\text{cl}}(o) < 0$; if none exist, correctness of predicting an empty set is evaluated.	0.10	2 s
Objects	Object-matching F1 combined with agreement in class, range, azimuth, velocity, and closing velocity for matched objects.	0.05	2 m, 5°, 1 m/s
Ego & command	$s_{\text{ego}} = \frac{3}{4}\frac{1}{2}\left[d(\lvert\Delta v\rvert; 0.5) + d(\lvert\Delta\omega\rvert; 2)\right] + \frac{1}{4}\mathbb{1}[\hat{c} = c]$.	0.05	0.5 m/s, 2°/s
Visual cue	Mean exact agreement over traffic-signal presence, red/green state, stop-line or crosswalk, and junction cues.	0.05	Exact
Object units	Fraction of object-level physical quantities expressed with the expected unit.	0.03	Exact
Coordinate units	Fraction of predicted trajectory coordinates expressed in the expected coordinate unit.	0.03	Exact
Rationale	Precision–recall agreement between physical quantities explicitly cited in the generated rationale and verifiable scene quantities derived from the annotations.	0.02	5%
3D box	Mean agreement in box center, dimensions, and heading for spatially matched objects.	0.01	2 m, 0.5 m

F Rationale Quality and Radar Grounding

We evaluate the structured physical rationales generated by *RaPhy-AD* on the same 1,287 trajectory anchors used in the controlled evaluation. We compare the rationale-supervised SFT model with the final SFT+GRPO model in terms of output validity, physical factuality, and rationale–trajectory consistency. We also test sensitivity to radar perturbations and assess the coverage of visual cues.

Rationale Validity and Physical Accuracy. Both models produce reliably structured outputs: the JSON parsing rate and complete-field rate are 0.998 for each model, and answer-unit accuracy changes from 0.999 with SFT to 1.000 with SFT+GRPO. Output formatting is therefore already nearly saturated after SFT.

Table 20 shows improvements in the accuracy of several physical quantities after GRPO. Ego $+1$ s speed MAE decreases from 0.212 to 0.126 m/s, object-range MAE from 1.033 to 0.762 m, object-velocity MAE from 0.816 to 0.654 m/s, and closing-velocity MAE from 0.762 to 0.712 m/s. Direction accuracy rises from 0.948 to 0.966, while the lead-vehicle detection rate rises from 0.733 to 0.756 and the no-object rate when a lead vehicle is present falls from 0.160 to 0.132. Command accuracy is unchanged at 0.998. Object F1 (0.340 versus 0.339) and hallucination rate (0.126 versus 0.127) change little; the gains in estimated physical quantities do not imply a corresponding improvement in every object-level metric.

Table 20: Quality of generated physical rationales on 1,287 evaluation anchors. Lower is better for MAE, no-object rate, hallucination rate, and incorrect-rationale/incorrect-trajectory rate; higher is better otherwise.

Metric	SFT	SFT+GRPO
Physical factuality		
Command accuracy	0.998	0.998
Ego +1s speed MAE (m/s)	0.212	**0.126**
Object range MAE (m)	1.033	**0.762**
Direction accuracy	0.948	**0.966**
Lead-vehicle detection rate	0.733	**0.756**
No-object rate w/ lead vehicle	0.160	**0.132**
Object velocity MAE (m/s)	0.816	**0.654**
Closing-velocity MAE (m/s)	0.762	**0.712**
Object F1	0.340	0.339
Object hallucination rate	0.126	0.127
Rationale–trajectory consistency		
Action matches future	0.750	**0.788**
Correct rationale & trajectory	0.650	**0.695**
Incorrect rationale & trajectory	0.122	**0.067**

Table 21: Radar grounding of the generated physical rationale for the SFT+GRPO model. Radar inputs are evaluated under original, mismatched, and zero-radar conditions.

Metric	Original	Mismatched	Zero
Object precision	**0.873**	0.676	0.795
Closing-velocity MAE (m/s)	**0.712**	3.440	1.957
Lead-vehicle range MAE (m)	**0.390**	0.789	0.687
Lead-vehicle detection rate	**0.756**	0.664	0.754

Rationale–Trajectory Consistency. GRPO also improves agreement between the generated rationale and the predicted trajectory (Table 20). The fraction of samples whose predicted action matches the actual future behavior increases from 0.750 to 0.788, and the fraction with both a correct rationale and a correct trajectory rises from 0.650 to 0.695. The fraction with both an incorrect rationale and an incorrect trajectory decreases from 0.122 to 0.067. Together, these results indicate better alignment between the generated reasoning and driving outcome on the evaluated anchors.

Radar Grounding of Physical Rationales. We compare original radar with mismatched radar from other clips and zero radar for the SFT+GRPO model (Table 21). Closing-velocity MAE increases from 0.712 m/s with original radar to 3.440 m/s with mismatched radar, a $4.8\times$ increase; zero radar yields 1.957 m/s. Relative to original radar, mismatched radar also raises lead-vehicle range MAE from 0.390 to 0.789 m, lowers object precision from 0.873 to 0.676, and lowers the lead-vehicle detection rate from 0.756 to 0.664. The corresponding zero-radar values are 0.687 m, 0.795, and 0.754. Mismatched radar is more damaging than zero radar on all four measures, consistent with the model relying on radar-derived quantities in its rationale and partially falling back on other observations when radar is absent.

Visual-Cue Coverage. Using Qwen3-VL-32B as a judge, we also evaluate whether the generated rationales mention visually observable scene elements. Precision for intersection, pedestrian/cyclist, construction/cone, and traffic-signal cues is 0.94, 0.83, 0.72, and 0.78, respectively, whereas recall is 0.43, 0.09, 0.15, and 0.51. The cues the model includes are relatively reliable, but the rationales do not cover every visible scene element.

F.1 Baseline Evaluation Details

All models are evaluated on the same trajectory anchors using a unified evaluation pipeline.

Common ground truth and coordinates. All predictions are evaluated against the same ground-truth ego trajectory, expressed in the ego frame at the anchor time (x forward, y left, with the origin at the rear axle), obtained from the offline ego-motion record. For baselines using a different coordinate convention, we estimate a 2D rotation between the model-specific and common ground-truth trajectories over the first $4\,\mathrm{s}$ and apply the same transformation to the corresponding prediction. Trajectories are resampled at $0.5\,\mathrm{s}$ intervals up to $3\,\mathrm{s}$, with linear interpolation applied when necessary, and all models are scored using the same evaluation code.

Route-command conditioning. RaPhy-AD, DriveWAM, and SimWAM receive route-conditioning information derived from the future ego trajectory, following their respective evaluation protocols. RaPhy-AD uses a LEFT/RIGHT/STRAIGHT command obtained from the ego yaw change over the subsequent $3\,\mathrm{s}$ using a $\pm 15^\circ$ threshold. DriveWAM incorporates route information through its scene-evolving guidance, while SimWAM uses the ground-truth endpoint direction following its official evaluation protocol. Alpamayo 1 does not expose an explicit navigation-conditioning interface. Alpamayo 1.5 supports navigation conditioning through `use_nav_prompt`; we use its official default configuration, in which this option is disabled.

DriveWAM guidance. DriveWAM conditions each rollout chunk on scene-evolving guidance. Since the public release provides guidance only for the clips used in the original work, we regenerate guidance for our evaluation clips using the pipeline provided by the official repository. We follow the same route-to-BEV-to-guidance procedure, guidance model, prompt format, image construction, resolution setting, and greedy decoding, while using `transformers` as the inference backend.

G Qualitative results

We provide qualitative comparisons between RaPhy-AD and Alpamayo 1.5 across diverse driving scenarios, including lane keeping and lane changes, interactions with vulnerable road users, dense traffic, and visually degraded conditions. While Alpamayo 1.5 generally produces concise driving rationales based primarily on visually inferred scene context, RaPhy-AD additionally grounds its reasoning in explicit physical quantities measured from radar, such as object range, azimuth, headway, and relative motion. The following examples illustrate how these physical cues provide additional evidence for interpreting surrounding agents and relating their motion states to the predicted ego trajectory.

Lane Keeping. Figure 4 shows a lane-keeping scenario in which the ego vehicle is approaching a lead vehicle while decelerating. RaPhy-AD explicitly identifies the vehicle at 39.5 m with a 1.7 s headway and a closing velocity of $-4.6\,\mathrm{m/s}$ as the primary constraint, directly connecting these measurements to its decision to slow down. In contrast, Alpamayo 1.5 focuses on a possible future lane merge, whereas RaPhy-AD explains the current longitudinal behavior using the measured state of the lead vehicle.

Lane Change. Figure 5 presents a highway scene with a heavy truck located to the left of the ego vehicle. RaPhy-AD uses both its lateral position and a positive closing velocity of $+11.4\,\mathrm{m/s}$ to recognize that the truck is moving away rather than posing an immediate longitudinal constraint, supporting continued acceleration. Alpamayo 1.5 instead provides the more generic rationale of maintaining distance from a lead vehicle without explicitly characterizing the truck's relative motion.

Pedestrian and Cyclist. Figure 6 shows a constrained roadway with a rider directly ahead of the ego vehicle. RaPhy-AD grounds its rationale in the rider's short 7.9 m range, near-zero relative velocity of -0.4 m/s, and position near the ego path, indicating that the rider is moving at a similar speed. While Alpamayo 1.5 also recognizes the need to maintain distance, RaPhy-AD provides explicit spatial and motion evidence supporting the same driving decision.

Cut-in. Figure 7 illustrates a nearby vehicle in an adjacent region of the highway that could be visually interpreted as a potential interaction. RaPhy-AD observes the vehicle at 21.0 m and uses its $+3.1$ m/s closing velocity to determine that it is pulling away, allowing the ego vehicle to continue accelerating. Alpamayo 1.5 instead gives a generic distance-maintenance rationale, without explicitly distinguishing whether the relative motion is increasing or decreasing the interaction risk.

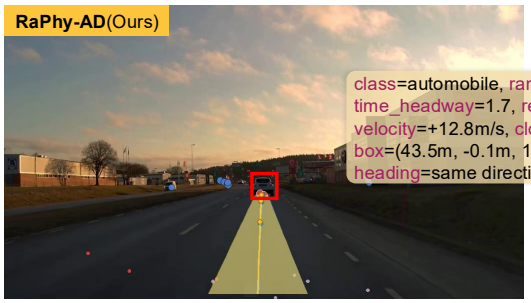
Dense Traffic. Figure 8 depicts a dense traffic scene in which the lead vehicle is only 8.8 m ahead. Despite the short range, RaPhy-AD observes a closing velocity of only -0.1 m/s, indicating that the ego and lead vehicle are nearly speed-matched, and accordingly maintains the current lane while slightly increasing speed. This demonstrates how combining distance with relative motion provides a more complete description of the interaction than reasoning from proximity alone.

Snow. Figure 9 presents a visually degraded snowy scene with a vehicle ahead of the ego vehicle. RaPhy-AD explicitly estimates the vehicle at 25.4 m with a closing velocity of -0.3 m/s, indicating near speed-matched motion and supporting stable lane following. The example highlights how radar-derived physical measurements can complement visual observations when scene appearance is degraded.

Fog. Figure 10 shows a foggy nighttime scene in which visibility is substantially reduced. RaPhy-AD identifies the lead vehicle at 26.0 m with a short 1.1 s headway and a closing velocity of -4.6 m/s, and therefore treats it as the primary constraint while slowing down. Although Alpamayo 1.5 also suggests adapting speed, RaPhy-AD explicitly grounds this decision in measured distance and relative motion.

Night. Figure 11 presents a nighttime scene containing a distant oncoming vehicle. RaPhy-AD recognizes that the vehicle is closing rapidly at -10.4 m/s but remains 54.5 m away, and therefore determines that it is not yet an immediate constraint on the ego trajectory. Alpamayo 1.5 instead focuses on the parked van on the right, whereas RaPhy-AD explicitly reasons about the motion state of the distant dynamic agent.

Lane Keeping



RaPhy-AD(Ours)

Visual cue: The ego is on a straight multi-lane road with a lead vehicle visible ahead and no ego-controlling traffic signals or junctions.

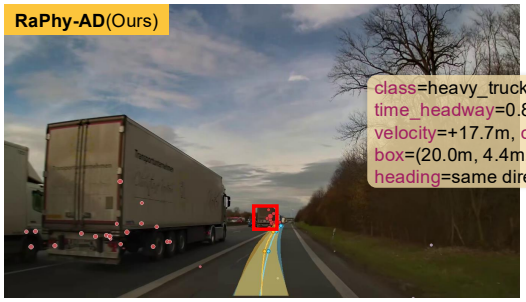
Rationale: With the ego decelerating from 17.7m/s at -0.5, the lead automobile 39.5m ahead at -0deg azimuth, closing at -4.6m/s with a 1.7s headway, is the primary constraint, so the ego continues straight while slowing. At that speed it covers about 18m per second.

Alpamayo

Rationale: Slow down to create a gap for merging left because our right lane tapers/ends ahead and a car is occupying the left lane, blocking an immediate merge while we maintain our lane behind the lead vehicle.

Figure 4: Qualitative comparison of RaPhy-AD and Alpamayo 1.5 in a lane-keeping scenario.

Lane Change



RaPhy-AD(Ours)

Visual cue: The ego is on a multi-lane highway with a heavy truck to its left and another vehicle further ahead, with no visible traffic signals or junctions controlling its lane.

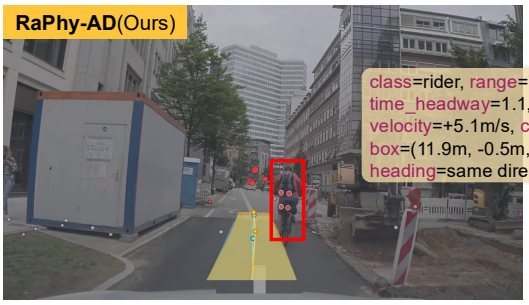
Rationale: With the ego accelerating from 14.2m/s at +0.7 and a yaw rate of -5.7deg/s, the heavy truck at 16.7m and +15.7 azimuth is moving away with a +11.4m/s closing velocity, so the ego continues accelerating while maintaining its lane on the highway. At that speed it covers about 14m per second.

Alpamayo

Rationale: Keep distance to the lead vehicle since it is directly ahead in our lane

Figure 5: Qualitative comparison of RaPhy-AD and Alpamayo 1.5 in a lane-change scenario.

Pedestrian and Cyclist



RaPhy-AD(Ours)

Visual cue: A green traffic signal controls the ego's lane, with construction barriers narrowing the roadway around a rider ahead.

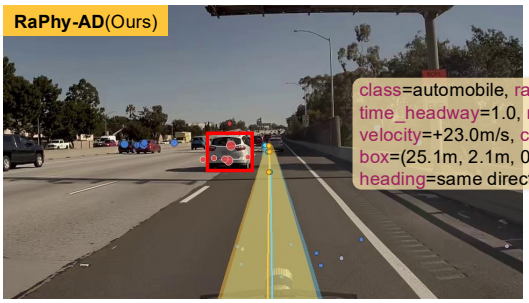
Rationale: With the ego accelerating from 5.6m/s at +0.6 and a green traffic signal ahead, the ego continues straight, maintaining its path behind the rider 7.9m ahead at -4deg azimuth, who is nearly speed-matched with a -0.4m/s closing velocity. At that speed it covers about 6m per second.

Alpamayo

Rationale: Keep distance to the lead cyclist since it is directly ahead in our lane

Figure 6: Qualitative comparison of RaPhy-AD and Alpamayo 1.5 in a pedestrian and cyclist scenario.

Cut in



RaPhy-AD(Ours)

Visual cue: The ego is on a multi-lane highway with an overhead sign indicating upcoming junctions, but no ego-controlling traffic signals or stop lines are visible.

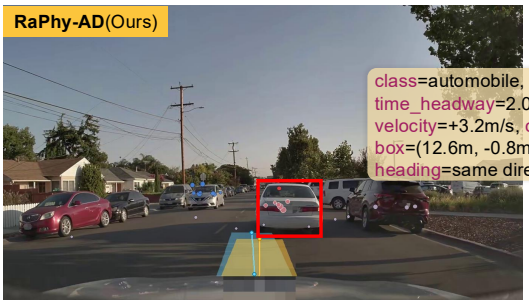
Rationale: With the ego accelerating from 20.1m at +0.6, and the automobile 21.0m ahead at +5.6m azimuth pulling away with a +3.1m/s closing velocity, the ego continues accelerating straight on the highway. At that speed it covers about 20m per second.

Alpamayo

Rationale: Keep distance to the lead vehicle since it is directly ahead in our lane

Figure 7: Qualitative comparison of RaPhy-AD and Alpamayo 1.5 in a cut-in scenario.

Dense Traffic



RaPhy-AD(Ours)

Visual cue: The ego is on a multi-lane road with parked vehicles on both sides and a line of traffic ahead, with no visible ego-controlling traffic signal or stop line.

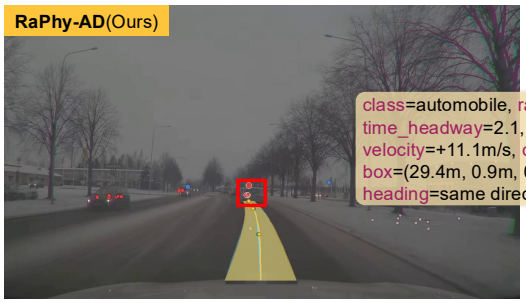
Rationale: With the ego moving at 3.3m/s and an automobile 8.8m ahead at -6deg azimuth nearly speed-matched with a closing velocity of -0.1m/s, the ego maintains its lane and continues straight, slightly increasing its speed. At that speed it covers about 3m per second.

Alpamayo

Rationale: Keep distance to the lead vehicle since it is directly ahead in our lane

Figure 8: Qualitative comparison of RaPhy-AD and Alpamayo 1.5 in dense traffic.

Snow



RaPhy-AD(Ours)

Visual cue: The ego is on a wet multi-lane road at dusk with streetlights on and other vehicles ahead, but no ego-controlling traffic signal or junction is visible.

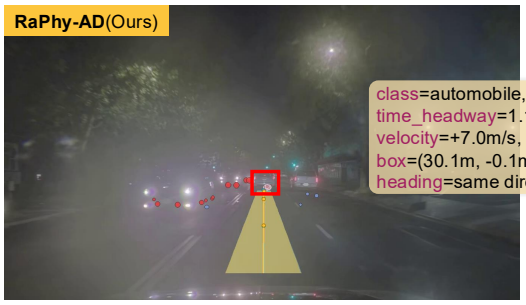
Rationale: With the ego maintaining its speed at 11.5m/s and slightly accelerating at +0.1, the automobile 25.4m ahead at +2.0deg azimuth is nearly speed-matched with a closing velocity of -0.3m/s, so the ego continues straight while maintaining its lane. At that speed it covers about 12m per second.

Alpamayo

Rationale: Keep distance to the lead vehicle since it is directly ahead in our lane

Figure 9: Qualitative comparison of RaPhy-AD and Alpamayo 1.5 under snowy conditions.

Fog



RaPhy-AD(Ours)

Visual cue: The ego is driving on a wet multi-lane road at night with several vehicles ahead and no visible ego-controlling traffic signal or junction.

Rationale: With the ego decelerating from 11.8m/s at -1.0, the automobile 26.0m ahead at -0deg azimuth with a 1.1s headway and closing at -4.6m/s is the primary constraint, so the ego continues straight while slowing down. At that speed it covers about 12m per second.

Alpamayo

Rationale:
Adapt speed to maintain a safe distance from the lead vehicle since it is directly ahead in our lane

Figure 10: Qualitative comparison of RaPhy-AD and Alpamayo 1.5 under foggy conditions.

Night



RaPhy-AD(Ours)

Visual cue: The ego is on a wet road at night with parked vehicles on the right and oncoming headlights visible in the distance, but no ego-controlling traffic signals or stop lines are visible.

Rationale: With the ego accelerating from 10.5m/s at +0.6 and maintaining a straight path, the automobile 54.5m ahead at +3.7deg azimuth, closing at -10.4m/s, is not an immediate constraint, so the ego continues straight and accelerates. At that speed it covers about 11m per second.

Alpamayo

Rationale: Nudge to the left to clear the stopped van blocking the right side of our lane

Figure 11: Qualitative comparison of RaPhy-AD and Alpamayo 1.5 under nighttime conditions.